FOUR-CLASS LEFT VENTRICULAR EJECTION FRACTION CLASSIFICATION FROM 12-LEAD ECG PRINTOUT IMAGES USING FOUNDATION MODELS, ENGINEERED ECG FEATURES AND PUBLIC-DATA TRANSFER (TRAIN100 EDITION)

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by

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APPROVAL OF THE PROJECT EVALUATION BOARD

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We certify that

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b. The work has not been submitted to any other Institute for any degree or diploma.

c. We have followed the guidelines provided by the Institute in writing the report.

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f. We took help from an AI assistant for some tasks, such as writing and checking code, running the documented experiments and checking the wording. All design decisions, data handling approvals and conclusions are our own.

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Abstract

Heart failure with reduced left ventricular ejection fraction (LVEF) is common and treatable but often detected late, because echocardiography is not available at every point of care. The 12-lead ECG is recorded routinely, yet in many Indian hospitals it survives only as a printed PDF page. This project grades LVEF into four classes (Normal, Mild, Moderate, Severe) from such printout images alone.

The data are 1,047 single-centre printouts from 993 patients, labelled from echocardiography: 837 ECGs for development and 210 locked until the final model was frozen. The traces were recovered exactly from the vector PDFs. Every model was evaluated with patient-grouped, repeated, nested cross-validation, corrected intervals and a written pre-registration. More than 1,300 results were recorded across ten stages, covering image CNNs, the ECGFounder and ECG-FM foundation models, LSTM and CNN-LSTM networks, 594 engineered features, gradient boosting, staged and ordinal ensembles and triage prediction. The public EchoNext data (4,827 ECGs) were redrawn in the hospital format and used for transfer learning.

The best four-class accuracy on all patients, from image-derived inputs only, is 0.587 [0.565, 0.610], above the model that also reads the printed machine measurements (0.562) and the original image CNN (0.471). A triage model that answers only confident cases is correct for 0.694 of the 56 % of patients it answers in cross-validation and 0.747 of the 50 % it answers on the locked test, scored once. Models trained only on EchoNext detect EF $\leq$ 45 % on the hospital printouts with AUROC 0.85. A simulation shows why all model families converge near 0.59: the ECG-based EF estimate correlates with echo at $r \approx 0.66$, and 0.70 on all patients would need $r \approx 0.90$. Phase II will use digital ECGs, larger linked datasets and verified Simpson-method EF labels.

In a final experiment (TRAIN100), every model that could be retrained without the EchoNext data was trained again, for as many epochs as needed, until its training

accuracy reached (nearly) 100 %. No model became reliably better on held-out patients: the main ensembles lost 2.6–3.8 points (for example IMG-C3 0.558 to 0.532), the triage model kept its accuracy (0.696) but answered fewer patients (48 % instead of 56 %), and eight of eleven networks on the fixed split lost accuracy, while the three that gained had been under-trained or gained within the noise of a 168-ECG split. This confirms that the early-stopped models, with training accuracy of about 0.60–0.85, are the right ones to use.

Index Terms: electrocardiogram, left ventricular ejection fraction, heart failure screening, ECG printout digitisation, ECG foundation model, ECGFounder, EchoNext, transfer learning, ensemble learning, selective prediction, nested cross-validation.

Contents

Certificate of Approval ii

Certificate iii

Declaration iv

Acknowledgements v

Abstract vi

List of Figures xi

List of Tables xiii

List of Abbreviations xv

1 Introduction 1

1.1 Background and motivation 1

1.2 Problem statement 1

1.3 Objectives 2

1.4 Contributions 2

1.5 Organisation of the report 3

2 Literature Review 4

2.1 Ejection fraction and its ECG signature 4

2.2 AI on the ECG for LV dysfunction 4

2.3 Foundation models and public data 4

2.4 Research gaps addressed in Phase I 6

3 Theoretical Study and Methodology 7

3.1 System overview 7
3.2 Dataset 8
3.3 Exact digitisation and the printout renderer 8
3.4 Input representations and features 8
3.5 Network architectures, preprocessing filters and training 9
3.5.1 Preprocessing filters 9
3.5.2 Network layers and activation functions 9
3.5.3 Training settings and training accuracy 9
3.5.4 Analysis of transfer learning 11
3.6 Evaluation protocol 13

4 Experimental Study **15**
4.1 Original project and Phase 1 16
4.2 Phases 2–5: robust evaluation, fusion and decision rules 17
4.3 Phases 6, 6A and 6B: fine-tuning and the LSTM family 17
4.4 Label-quality, cohort and abstention studies 18
4.5 Phase 7 and the image-only ensembles 19

5 Public-Data Transfer and Image-Only Deep Learning **21**
5.1 EchoNext in the printout format 21
5.2 Pruned ensembles of all image-derived models 22
5.3 Additional features for Mild and Moderate 23
5.4 Image-only pixel models with EchoNext pre-training 23

6 Results and Discussion **26**
6.1 Headline results 26
6.1.1 Best on all patients: 0.587 27
6.1.2 Best in the 60s: 0.694 on the answered patients 27
6.1.3 Best in the 70s: 0.747 on unseen patients 28
6.2 All models compared 29
6.3 All results in the upper 50s, 60s and 70s 30
6.4 Trade-off analysis 30
6.5 Why the results are strong and why higher accuracy is not feasible . . 34

7 Training Every Model to 100 % Training Accuracy **36**

7.1 Method 36

7.2 Development cross-validation models 37

7.3 Image and signal networks on the fixed split 37

7.4 Variants of every phase 38

7.5 Label-quality studies 40

7.6 Models that were not retrained 43

7.7 Why training to 100 % does not help, and where to stop 43

8 Summary and Conclusions **46**

8.1 Conclusions 46

8.2 Scope for future work (Phase II) 47

References **48**

List of Figures

3.1 System architecture of the Phase I pipeline. Dashed arrows: knowledge transferred from the EchoNext public dataset. 7

3.2 Training curves of six representative networks: training and validation accuracy (solid) and loss (dotted) per epoch. 11

4.1 Phase 6B architectures: waveform encoders (A), early fusion with the header and ECGFounder (B) and late fusion with the vote ensemble (C). 19

5.1 Four-class accuracy of every stack and ensemble variant of the public-data study, with corrected 95 % confidence intervals. 24

6.1 Left: development estimates and locked-test results of the final model (every cross-validation value lies inside its locked-test interval). Right: locked-test confusion matrix of the four-class image-only classifier. . . 30

6.2 All model families sorted by four-class accuracy (highest at the top). [test] = fixed 210-ECG test set of the original project; [CV] = cross-validation on 837 development ECGs; [locked test] = frozen model scored once. 31

6.3 The same models sorted by heart-failure screen AUC (reduced EF against Normal). 32

6.4 Every result with accuracy ≥ 0.55, grouped by evaluation population (each dot is one configuration; stars mark the headline results). . . . 33

6.5 Accuracy against the share of ECGs answered (selective prediction, nested thresholds). 33

6.6 Mild detected (recall of the Mild class) against four-class accuracy for all models. Stars mark the headline results; lines show the effect of a Mild offset on IMG-vote (all ECGs) and on the triage model (answered ECGs). 34

6.7 Expected four-class accuracy against the SD of the EF estimation error (simulation on the 837 development ECGs). Our EF estimators lie at SD 13–14, where the best model (0.587) sits on the curve. 70 % needs SD $\leq$ 7.7, i.e. $r \approx 0.90$. 35

6.8 Error rate by distance of the echo EF from the nearest class limit. . . 35

7.1 Development cross-validation: training accuracy after training to (nearly) 100 % (blue), held-out accuracy of the same model with its corrected 95 % interval (orange) and held-out accuracy of the original early-stopped model (green). 39

7.2 Original fixed split: training accuracy after training to $\geq$ 99.5 % (blue), validation accuracy of the same network (orange) and validation accuracy of the original network at its selected epoch (green). 42

7.3 ECGFounder MLP head trained for 150 epochs without early stopping (mean of the five folds of repeat 0): training accuracy reaches 1.000 while held-out accuracy stays at about 0.52. Left: the project head; right: the same head without dropout or weight decay. The dashed line marks the early-stopping epoch. 45

List of Tables

2.1 Relevant studies: data, input, task and main reported result (as printed in each abstract). 5

3.1 Preprocessing filters and their parameters for each input representation. 10

3.2 Layers, activation functions, normalisation and size of every network used in Phase I. 12

3.3 Training against validation accuracy of the main networks. Upper part: original project, fixed training/validation split, at the selected epoch (training accuracy is measured during the epoch, with augmentation and dropout active). Lower part: frozen-feature models in development cross-validation (repeat 0, seed 42): accuracy on the training folds against the held-out fold. 13

3.4 Transfer learning: source, how the knowledge was transferred and what it gave on the hospital printouts. 14

4.1 Overview of the experimental stages of Phase I. 15

4.2 Representative models of the original project (fixed test set, $n = 210$). 16

4.3 Phase 1 key models (fixed test set). 16

4.4 Phase 5, development cross-validation (min recall = lowest per-class recall; admissible if ≥ 0.15). 17

4.5 LSTM family: architecture, size, cost and results (development CV, original labels). 18

4.6 LSTM fusions (development CV, original labels). 18

4.7 Label-quality, cohort and abstention studies (development CV). . . . 19

4.8 Image-only four-class models (all 837 ECGs, development CV). 20

5.1 EchoNext-trained models: internal checks and transfer to the 837 hospital ECGs without any hospital training. 21

5.2 All ensemble variants (837 ECGs, original labels, corrected 95 % CI). Members are the mean kept per fold. 22

5.3 Pixel models on the 837 hospital images (class weights tuned inside the training folds). 25

6.1 Headline results of Phase I (image-derived inputs only; original four-class labels), with the result of the same model trained to (nearly) 100 % training accuracy where it could be retrained. 26

7.1 What changed in TRAIN100. Inputs, labels, architectures, folds and decision rules are the original ones; only the brakes on fitting were removed, and every model was trained for as many epochs or iterations as it needed. 37

7.2 Development cross-validation (5 folds × 3 repeats, 837 ECGs): every model trained to (nearly) 100 % training accuracy (TRAIN100; training and held-out columns) against the original early-stopped model. Held-out 95 % CI corrected for repeated CV; change = TRAIN100 minus original, in percentage points. 38

7.3 Image and signal networks on the original fixed split (trained on 669 ECGs, measured on 168 validation ECGs; the locked test is not loaded). TRAIN100: no early stopping, augmentation, weight decay, label smoothing or EMA, trained until training accuracy ≥ 99.5 %. Original training accuracy was measured during the epoch with augmentation and dropout active. 40

7.4 Variants of every phase, trained to (nearly) 100 % training accuracy in the same development cross-validation. Members without an original stand-alone result show –. 41

7.5 EF regression (Phase 4) trained without early stopping: correlation with echo EF on the training rows and held out, and mean absolute error in EF points. 42

7.6 Label-quality studies (report Table 4.7) with the C3 recipe trained to (nearly) 100 % training accuracy, scored on the same subsets as the original. 43

7.7 Models not retrained in TRAIN100, and why. 44

List of Abbreviations

AUROC	Area under the receiver operating characteristic curve
CI	Confidence interval (corrected for repeated cross-validation unless stated)
CNN	Convolutional neural network
CV	Cross-validation
ECG	Electrocardiogram
ECGF, ECGFounder	ECG foundation model pre-trained on more than 10 million ECGs (Li et al.)
EchoNext	Public ECG–echocardiography dataset (PhysioNet), about 100,000 ECGs
EF, LVEF	(Left ventricular) ejection fraction, in per cent
GBM	Gradient-boosting machine (LightGBM)
HF	Heart failure
LR	Logistic regression
LSTM	Long short-term memory recurrent network
Mi, Mo, N, S	Mild, Moderate, Normal and Severe classes
MLP	Multi-layer perceptron
STAGED-4	Staged four-class decision: Normal / {Mild, Moderate} / Severe, then Mild vs Moderate
SUPER	Ensemble of all image-derived members pruned inside each training fold
TCRT	Total cosine R-to-T
TOP10	Ensemble keeping the ten best single members per fold

Notation

y	Class label $y \in \{0, 1, 2, 3\}$: Normal, Mild, Moderate, Severe
$\hat{\mu}$	Predicted EF (per cent)
σ	Standard deviation of the EF estimation error
r	Pearson correlation between predicted and echo EF
$\Phi(\cdot)$	Standard normal cumulative distribution function
$\mathbf{w}$	Class weights of the decision rule
J	Number of cross-validation cells (here 15)

Chapter 1

Introduction

1.1 Background and motivation

Heart failure affects tens of millions of people, and its outcome depends on how early left ventricular (LV) systolic dysfunction is recognised [1, 2]. Systolic function is graded by the left ventricular ejection fraction (LVEF), measured by echocardiography, which needs an ultrasound machine and a trained operator. The 12-lead electrocardiogram (ECG), in contrast, is recorded for almost every cardiac patient, and deep learning has shown that it carries information about reduced LVEF [3, 4]. That evidence comes mostly from digital signals and very large datasets. In many Indian hospitals the ECG survives only as a printed page or its PDF, which shows each lead for just 2.5 s plus one 10-s rhythm strip.

1.2 Problem statement

Given only the image of a hospital 12-lead ECG printout, predict the patient's four-class LVEF category:

- **Normal:** EF above 52 % (men) or at least 54 % (women) [5];
- **Mild:** 41 % up to the Normal limit;
- **Moderate:** 31 to below 41 %;
- **Severe:** below 31 %.

The model may use only the waveform on the page: no printed machine measurements, age or sex. The Mild band, only 11–13 EF points wide, is where an early warning is most valuable and where the ECG changes are subtle. A useful model must therefore detect Mild and not only the extremes.

1.3 Objectives

1. Recover the exact ECG signal from the printout and build image-derived representations of it.
2. Train and compare the widest practical range of models, from classical features and CNNs to foundation models, LSTMs and ensembles, under a nested and pre-registered protocol.
3. Transfer knowledge from public ECG–echo data (EchoNext) drawn in the hospital printout format.
4. Find the best all-patient four-class model and a triage model that answers confidently, and confirm it once on a locked test set.
5. Quantify the accuracy that the data can support.

1.4 Contributions

- An exact vector digitiser and an inverse printout renderer for the hospital ECG format.
- More than 1,300 evaluated configurations across ten experimental stages, all documented.
- A best all-patient four-class accuracy of **0.587** from image-derived inputs only, a triage accuracy of **0.694** on 56 % of patients, and **0.747** on unseen locked-test patients.
- A quantitative explanation of the accuracy ceiling in terms of the precision of the EF labels.

1.5 Organisation of the report

Chapter 2 reviews the literature. Chapter 3 describes the data, digitisation, features, networks and evaluation protocol. Chapter 4 reports the experimental study phase by phase and Chapter 5 the public-data transfer. Chapter 6 presents the results and their analysis. Chapter 7 retrains every available model to (nearly) 100 % training accuracy and compares it with the original, and Chapter 8 concludes with the plan for Phase II.

Chapter 2

Literature Review

2.1 Ejection fraction and its ECG signature

Guidelines grade LVEF into normal, mildly reduced and reduced ranges [1, 2], and echocardiography quantifies it [5]. Repeat measurements of the same patient can differ by about 10 EF points [6], which matters when classes are only 10–13 points wide. On the ECG, LV dysfunction appears as QRS prolongation and bundle-branch block [7], fragmented QRS [8], a wide spatial QRS–T angle [9] and repolarisation heterogeneity [10, 11].

2.2 AI on the ECG for LV dysfunction

Convolutional networks on digital 12-lead ECGs detect EF $\leq$ 35 % with AUC 0.93 [3], and a pragmatic trial raised new diagnoses in primary care [4]. Graded estimation is less common: Vaid et al. predicted three EF bands and regressed EF from 715,890 ECGs [12]. Models on ECG *images* reach AUROC 0.90–0.95 for EF $<$ 40 % when trained on 385,601 ECGs [13], and image digitisation is an active research area [14–16]. Composite structural-disease models showed that each additional input type raises discrimination [17], and single-lead and wearable studies extended screening beyond the hospital [18–20].

2.3 Foundation models and public data

Table 2.1 Relevant studies: data, input, task and main reported result (as printed in each abstract).

Study	Data	Input	Task	Main reported result
Attia et al. [3]	44,959 train / 52,870 test patients, Mayo Clinic	digital 12-lead	EF $\leq$ 35 %	AUC 0.93, accuracy 85.7 %
Attia et al. [32]	3,874 prospective patients	digital 12-lead	EF $\leq$ 35 %	AUC 0.918; 39.8 % of false positives had EF 36–50 %
Yao et al. [4]	22,641 adults, 120 care teams (RCT)	digital 12-lead	new low-EF diagnosis	diagnoses 1.6 % $\rightarrow$ 2.1 % (OR 1.32)
Adedinsewo et al. [33]	1,606 emergency patients	digital 12-lead	EF $\leq$ 35 % / $<$ 50 %	AUROC 0.89 / 0.85
Vaid et al. [12]	715,890 ECGs, 147,636 patients, 5 hospitals	digital 12-lead	EF 3 classes + regression	AUC 0.94 / 0.82 / 0.89 (internal); MAE 5.8–6.1
Kwon et al. [34]	55,163 ECGs, 2 hospitals	ECG features + demographics	HFrEF (EF $\leq$ 40 %)	AUROC 0.843 internal, 0.889 external
Sangha et al. [35]	2.2 million ECGs plotted as images	ECG images	6 rhythm / conduction labels	mean AUROC 0.99 (external 0.97)
Sangha et al. [13]	385,601 ECGs, Yale + external sites	ECG images	EF $<$ 40 %	AUROC 0.91 internal, 0.90–0.95 external
Ulloa-Cerna et al. [17]	2.2 million ECGs, 484,765 adults	ECG + age, sex	composite structural disease	AUROC 0.91
Elias et al. [36]	77,163 patients	digital 12-lead	moderate / severe AS, AR, MR	AUROC 0.88 / 0.77 / 0.83
Poterucha et al. [23]	$>$ 1 million ECG–imaging pairs	digital 12-lead	structural heart disease	outperformed cardiologists; data and weights released
Hughes et al. [24]	100,000 ECGs, 36,286 patients (public)	digital, 250 Hz	composite + 11 labels	composite AUROC 82.0
McKeen et al. [22]	1.5 million ECGs (pre-training)	digital 12-lead	EF $\leq$ 40 % and others	AUROC 0.929 for EF $\leq$ 40 %
Li et al. [21]	10.8 million ECGs, 150 labels	digital 12-lead	broad diagnosis	AUROC $>$ 0.95 for 80 diagnoses

Table 2.1 (continued)

Study	Data	Input	Task	Main reported result
This work	837 development + 210 locked printouts, one centre; EchoNext 4,827	printout images (and derived traces)	**four-class EF**	0.587 all patients; 0.694 (56 % answered); 0.747 locked test (50 %); EF $\leq$ 45 % AUROC 0.85 from EchoNext only

ECGFounder was pre-trained on 10.8 million ECGs and transfers to new tasks with little data [21]. ECG-FM reported AUROC 0.929 for EF $\leq$ 40 % [22]. EchoNext released ECG–echocardiography pairs and models that outperformed cardiologists [23, 24]. Earlier 1-D networks reached cardiologist-level rhythm classification [25] and automated 12-lead diagnosis [26]. The image backbones used in this project are ResNet [27], EfficientNet [28], MobileNetV3 [29] and ConvNeXt [30]; the sequence models are the LSTM [31] and its convolutional variants.

2.4 Research gaps addressed in Phase I

- **Graded EF from printouts.** Published work is mostly binary and uses digital signals; four-class grading including Mild from printout images had not been reported.
- **Small single-centre data.** Methods for a few hundred labelled printouts, with nested evaluation, are rarely studied.
- **Transfer to the printout format.** Public ECG–echo data had not been re-drawn in a hospital's printout format for pre-training.
- **Confident subsets.** Triage that answers only confident cases and refers the rest is seldom evaluated for graded EF.

Chapter 3

Theoretical Study and Methodology

3.1 System overview

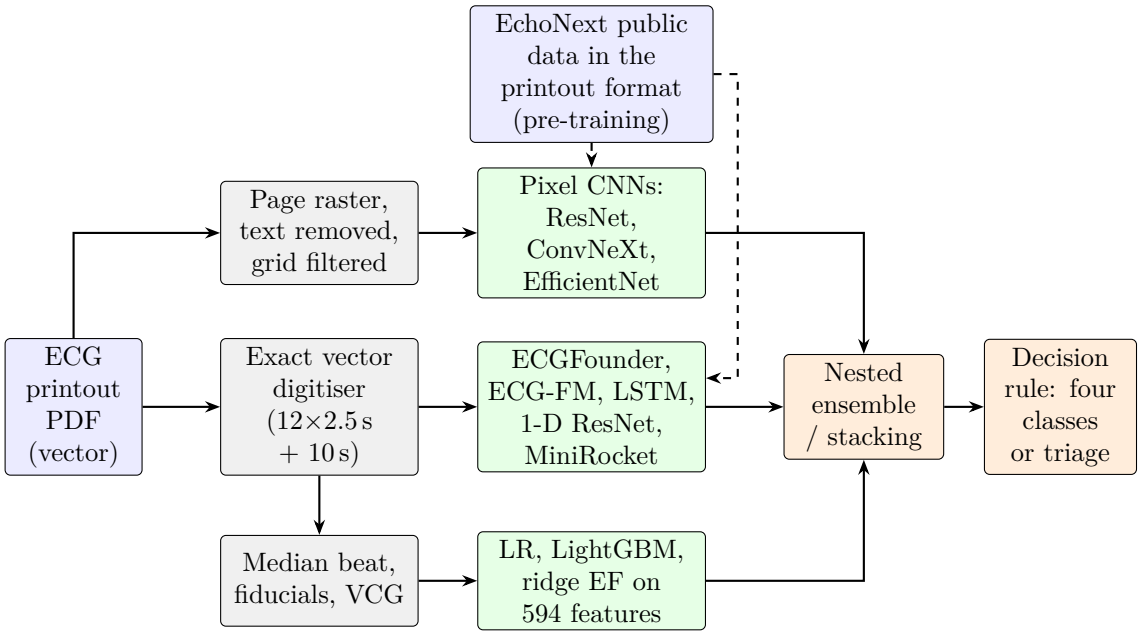


Figure 3.1 System architecture of the Phase I pipeline. Dashed arrows: knowledge transferred from the EchoNext public dataset.

The Phase I pipeline (Figure 3.1) has three parts. The printout PDF is digitised exactly, turned into image, signal and feature representations, and passed to the model families. External knowledge enters from the public EchoNext data, redrawn in the hospital printout format.

3.2 Dataset

The data are 1,047 12-lead ECG printouts (PDF) from 993 patients of one tertiary hospital, recorded in 2020–2024. Each page has the standard 3 × 4 layout (2.5 s per lead), a 10-s lead II rhythm strip, 25 mm/s and 10 mm/mV. Each ECG was matched to an echocardiography report of the same patient, whose EF gives the class. The data were split once by patient into a **develop- ment set** of 837 ECGs (384 Normal, 185 Mild, 177 Moderate, 91 Severe) used for all modelling and a **locked test set** of 210 ECGs (96 / 47 / 44 / 23) held back until the final model was frozen.

3.3 Exact digitisation and the printout renderer

The PDFs are vector drawings: each lead is a polyline of about 700 segments, so the trace is recovered exactly. With the row baselines y_{base} and column origins x_0 of the page,

$$v(t) = \frac{y_{\text{base}} - y(x)}{82}\,\text{mV}, \qquad t = \frac{x - x_0}{205.2}\,\text{s}, \tag{3.1}$$

resampled to 500 Hz. The inverse operation, a *printout renderer*, draws any 12-lead signal on a page identical to the hospital's, rounding each point to whole drawing units as the machine does. Redrawn development ECGs match their real pages: 81 % of drawn points are identical and the read-back error is 5–13 μ V.

3.4 Input representations and features

Four representations were built from each printout:

- **page image**: the waveform area with text removed and the grid filtered, for CNNs;
- **aligned signal**: the 12 leads in their own 2.5-s windows on a common 10-s axis, for ECGFounder and the LSTMs;
- **ECGFounder embeddings**: the 1,024-d final embedding and 150 diagnostic outputs;

- **engineered features**: 594 features, among them 191 clinician-style and vectorcardiographic measurements such as the spatial QRS–T angle, total cosine R-to-T (TCRT) and the QT variability index.

3.5 Network architectures, preprocessing filters and training

3.5.1 Preprocessing filters

Each representation has its own filter chain (Table 3.1): a colour filter removes the pink grid from page images, and a 50 Hz notch, a 0.67–40 Hz band-pass and a moving-median baseline filter clean the signals.

3.5.2 Network layers and activation functions

Table 3.2 gives the layers, kernels, activation functions, normalisation and size of every network. The activation functions used are

$$\begin{gathered}\mathrm{ReLU}(x)=\max(0,x), \qquad \mathrm{SiLU}(x)=x\,\sigma(x), \qquad \mathrm{GELU}(x)=x\,\Phi(x),\\ \text{h-swish}(x)=x\,\frac{\mathrm{ReLU6}(x+3)}{6}, \qquad \sigma(x)=\frac{1}{1+e^{-x}},\\ \mathrm{softmax}(\mathbf{z})_c=\frac{e^{z_c}}{\sum_{j=1}^{4}e^{z_j}},\end{gathered} \tag{3.2}$$

where Φ is the standard normal distribution function. Every four-class network ends in a softmax; ordinal heads use sigmoids for $P(y>k)$, and LSTM gates use the sigmoid with a tanh cell activation.

3.5.3 Training settings and training accuracy

Image CNNs were trained with AdamW (learning rate 10^{-3}, weight decay 0.05), warm-up and cosine decay, batch 16, up to 30 epochs with early stopping, class-weighted cross-entropy with label smoothing 0.05, weight averaging (EMA 0.99) and augmentation (rotation $\pm 2°$, time and amplitude scaling, translation, random erasing). ECG-Founder was fine-tuned with differential learning rates (backbone 10^{-4}, head 10^{-3}).

Table 3.1 Preprocessing filters and their parameters for each input representation.

Input	Step	Filter / operation and parameters
Page image (CNNs)	rasterise and crop	vector PDF rendered at 1 px/pt; crop to the waveform area (removes all header text)
	grid removal	colour filter: darkness $= 255 - \max(R, G, B)$ (HSV value); the pink grid has a high value and vanishes
	binarisation	Otsu threshold (clamped)
	denoising	connected components < 6 px removed (keeps 1-px trace lines that a median filter would erode)
	thickening, resizing	scale-aware dilation; area interpolation to 512×1024, then 256×512 network input
	filter ablations	Gaussian 5×5, $\sigma = 1$; median 3×3; bilateral (d 7, σ 50/7); CLAHE (clip 2.0, 8×8 tiles); adaptive Gaussian threshold (block 25, C 60)
Digitised signal (ECGFounder)	exact digitisation	vector paths converted at 82 pt/mV and 616/3 pt/s; 500 Hz
	power-line notch	IIR notch at 50 Hz, $Q = 30$
	band-pass	4th-order Butterworth, 0.67–40 Hz, zerophase (forward–backward)
	baseline	0.4-s moving-median baseline subtracted
	scaling, layout	z-score per lead; 12 leads in their own 2.5-s windows on a 10-s axis ($12 \times 5{,}000$)
LSTM / CNN-LSTM	as above, then decimation	100 Hz; amplitudes clipped at ± 5 mV; 13 signals + 13 mask channels ($26 \times 1{,}000$)
Engineered features	beat analysis	0.5 Hz high-pass; R-peak detection; Raligned median beats; vectorcardiogram by the Kors matrix
EchoNext signals	harmonisation	de-normalisation to μ V; resampling $250 \rightarrow 500$ Hz; cascaded median filters (0.2 s, 0.6 s) for baseline removal; clipping; printout cut (2.5-s windows)
Pixel CNNs (IMG)	page image	waveform area, text removed, grid filtered; 320×640 (ResNet-18) or 384×768 (ConvNeXt-Tiny, EfficientNet-B2); grey channel repeated to RGB

LSTMs used AdamW (10^{-3}), batch 32, gradient clipping and early stopping on inner validation. Table 3.3 compares training and validation accuracy.

- **Image CNNs** reach 0.52 on training and 0.48–0.52 on validation data: they learn evenly and are not over-fitted.
- **Foundation-feature heads** reach 0.73–0.76 on training folds against 0.51 held out, and fine-tuning drives the training loss to 0.66 while validation loss stays at 1.17. With 837 ECGs for 30.8 million parameters, frozen features, regularisation, nested selection and ensembling were therefore used in all later phases.

3.5.4 Analysis of transfer learning

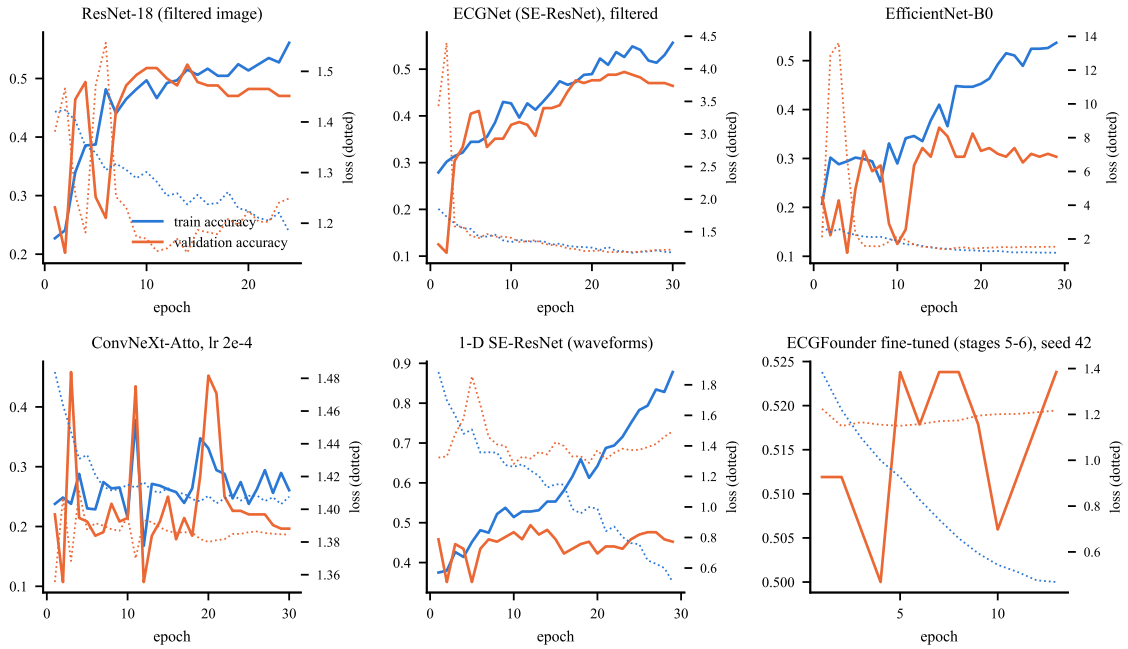


Figure 3.2 Training curves of six representative networks: training and validation accuracy (solid) and loss (dotted) per epoch.

Transfer learning was used in four forms, from natural images to EF-labelled ECGs (Table 3.4). The closer the source is to the target, the larger the gain: ECG pre-training (ECGFounder) beat ImageNet pre-training by 4–5 points, and EF-labelled public ECGs (EchoNext) supplied the knowledge behind the best ensembles. A model trained only on US patients separates reduced from normal EF on our printouts with AUROC 0.85, which also validates the printout conversion. Frozen ECGFounder features match fine-tuning in EF correlation (0.652 against 0.650) without its over-

Table 3.2 Layers, activation functions, normalisation and size of every network used in Phase I.

Network	Layers and blocks	Activation	Norm.	Params
ResNet-18	7×7 conv stem + max-pool; 4 stages $\times$ 2 basic blocks of two 3×3 convs (64–512 channels); global average pool; FC(4) — 18 weight layers	ReLU	BatchNorm	11.2 M
ECGNet (custom)	$7 \times 7/2$ stem; 4 stages of 1–1–2–2 residual SE blocks (widths 64–256) with time-elongated 3×5 kernels; [GAP ‖ GMP]; dropout 0.3; FC(4) —13 conv layers	ReLU; sigmoid (SE)	BatchNorm	6.3 M
EfficientNet-B0	3×3 stem; 16 MBConv blocks in 7 stages with SE; 1×1 head conv; FC(4)	SiLU	BatchNorm	4.0 M
EfficientNet-B2 (IMG70)	3×3 stem; 23 MBConv blocks in 7 stages with SE; 1×1 head conv; FC(4) + EF head	SiLU	BatchNorm	7.7 M
MobileNetV3-Large	3×3 stem; 15 inverted-residual bottlenecks, SE in 8; 1×1 head	h-swish, ReLU	BatchNorm	4.2 M
ConvNeXt-Atto	4×4 patchify stem; 4 stages of 2–2–6–2 blocks (7×7 depth-wise conv + MLP)	GELU	LayerNorm	3.4 M
ConvNeXt-Tiny (IMG70)	4×4 patchify stem; 4 stages of 3–3–9–3 blocks (96–768 channels); FC(4) + EF head	GELU	LayerNorm	27.8 M
1-D SE-ResNet	1-D residual blocks, kernel 9, SE; [GAP ‖ GMP]; FC(4)	ReLU	BatchNorm	–
ECGFounder (Net1D)	1-D conv stem (kernel 16, stride 2); 7 stages of 2–2–2–3–3–4–4 blocks (20 blocks; 64–1,024 filters), grouped kernel-16 convs with SE; GAP $\rightarrow$ 1,024-d embedding	Swish	none	30.8 M
MLP head / fusion	dropout 0.5 $\rightarrow$ linear 64 $\rightarrow$ GELU $\rightarrow$ dropout 0.5 $\rightarrow$ linear 4	GELU	standardised input	66 k
LSTM L0–L4	1–2 LSTM layers, 64 hidden units, uni- or bidirectional; [last state ‖ masked mean]; dropout 0.3; linear 4	sigmoid, tanh	–	18–147 k
CNN-LSTM (L5)	$3 \times$ [Conv1d k 7, stride 2] ($26 \rightarrow 32 \rightarrow 64 \rightarrow 64$ channels); BiLSTM-64; linear 4	GELU; sigmoid, tanh	–	117 k
Meta-learners	multinomial logistic regression; Frank–Hall ordinal (3 binary logistic models)	softmax; sigmoid	standardised	< 1 k

Table 3.3 Training against validation accuracy of the main networks. Upper part: original project, fixed training/validation split, at the selected epoch (training accuracy is measured during the epoch, with augmentation and dropout active). Lower part: frozen-feature models in development cross-validation (repeat 0, seed 42): accuracy on the training folds against the held-out fold.

Network	**Epochs (selected)**	**Training accuracy**	**Validation / held-out accuracy**	**Train / val loss**
ResNet-18 (filtered image)	24 (14)	0.515	0.524	1.237 / 1.142
ECGNet (SE-ResNet)	30 (21)	0.523	0.476	1.222 / 1.213
EfficientNet-B0	29 (19)	0.447	0.351	1.327 / 1.436
ConvNeXt-Atto (10^{-3})	30 (25)	0.404	0.452	1.336 / 1.318
1-D SE-ResNet on digitised waveforms	29 (14)	0.553	0.482	1.127 / 1.344
ECGFounder fine-tuned (stages 5–6)	13 (8)	–	0.524	0.660 / 1.174
ECGFounder linear head (CV)	–	0.730	0.510	–
ECGFounder MLP head (CV)	–	0.761	0.514	–
Header + ECGFounder MLP fusion (CV)	–	0.668	0.535	–
Header + clinical features, logistic regression (CV)	–	0.562	0.507	–

fitting, and pre-training a pixel CNN on EchoNext ECGs drawn as hospital printouts adds 3.8 points over ImageNet initialisation.

3.6 Evaluation protocol

- **Cross-validation:** patient-grouped 5-fold CV repeated three times (15 cells), identical folds for every model.
- **Nested choices:** every hyper-parameter, class weight, threshold and ensemble member is chosen inside the training folds [37]. Class weights maximise accuracy subject to every recall ≥ 0.18 (fallback 0.15):

$$\hat{y} = \arg\max_k \; w_k P_k, \qquad \mathbf{w}^* = \arg\max_{\mathbf{w}} \; \mathrm{Acc}(\mathbf{w}) \;\; \text{s.t.} \;\; \min_k \mathrm{Rec}_k(\mathbf{w}) \geq 0.18. \quad (3.3)$$

Table 3.4 Transfer learning: source, how the knowledge was transferred and what it gave on the hospital printouts.

Source	Transfer method	Result on the hospital data
ImageNet (1.3 M natural images)	initialise ResNet-18; fine-tune on page images	0.492 four-class (CV); natural-image features transfer only partly to thin line drawings
ECGFounder (10.8 M ECGs, 150 labels)	frozen 1,024-d embedding + linear or MLP head	0.528–0.542 (CV); screen AUC 0.82; EF r 0.65; best single representation of the project
	partial fine-tuning (stages 5–6; stage 6)	0.536 (CV); EF r 0.650 against 0.652 frozen: the frozen embedding already holds the EF information
	layer choice (stage 4, 5, 6, diagnostic outputs)	final embedding best (0.544 against 0.517–0.531)
EchoNext signals (4,827 US ECG– echo pairs)	models trained only on EchoNext, applied zero-shot	AUROC 0.85 for EF $\leq$ 45 %, 0.75 Normal against Mild, EF r 0.62; four-class 0.551 with no hospital training of the model (only its decision thresholds tuned)
	EchoNext models as stack members	SUPER ensembles 0.583–0.587, the best all-patient results
EchoNext images (drawn as printouts)	pre-train pixel CNN, then fine-tune	final ConvNeXt-Tiny ensemble 0.548; ResNet-18: 0.530 against 0.492 with ImageNet only (+3.8 points); EchoNext test accuracy of the pre-trained CNNs 0.58–0.72

- **Intervals and tests:** corrected resampled t statistic [38] with Holm adjustment [39]:

$$t = \bar{d} \Big/ \sqrt{\left(\frac{1}{J} + \frac{n_2}{n_1}\right)\hat{\sigma}_d^2}, \qquad J = 15. \tag{3.4}$$

- **Integrity:** every study was pre-registered in a dated document before it was run; the locked test set was scored once.

Chapter 4

Experimental Study

Table 4.1 Overview of the experimental stages of Phase I.

Stage	**Main question**	**Evaluation**	**Outcome carried forward**
Original project	Can a page-image CNN grade LVEF?	fixed 210-ECG test set	27 models; header + ResNet-18 fusion 0.552
Phase 1	Does a foundation ECG model help?	fixed test set	ECGFounder best for Mild (recall 0.26)
Phase 2	Robust re-evaluation, pre-registered	development CV	ECGFounder MLP fusion 0.544
Phase 3	Fusion, ensembles, calibration	development CV	vote ensemble C1 0.547
Phase 4	EF regression, multitask, label quality	development CV	EF $r = 0.66$; information limit found
Phase 5	Features, boosting, decision rules	development CV	C3 selected, 0.562
Phase 6 / 6A / 6B	Fine-tuning, LSTM family, label variants	development CV	recurrent models tested; C3 kept
Label studies	Cleaning, cohorts, abstention	CV and subsets	selective prediction reaches 0.71
Phase 7	Representations, learning curve, triage	CV, then locked test	triage 0.694 (CV), 0.747 (locked test)
Image-only push	HIER, ENH, ULT, EchoNext, SUPER	development CV	ensembles 0.57–0.587

Phase I was carried out in ten stages(Table 4.1). Each stage asked one question, and its lesson was carried into the next. Together they produced more than 1,300 recorded results; the Supplementary Tables list every configuration, and Figure 6.2 in Chapter 6 compares all model families.

4.1 Original project and Phase 1

Table 4.2 Representative models of the original project (fixed test set, $n = 210$).

Model	Accuracy	Bal. acc.	Macro-F1	QWK	HF AUC
HOG + PCA + RBF-SVM (classical)	0.481	0.322	0.283	0.229	0.737
ECGNet (custom SE-ResNet)	0.471	0.360	0.262	0.389	0.745
ResNet-18, ImageNet initialised	0.486	0.356	0.323	0.290	0.735
EfficientNet-B0 / MobileNetV3-L (from scratch)	0.300 / 0.143	0.318 / 0.243	0.236 / 0.111	–	0.712 / 0.649
1-D ResNet on digitised waveforms (×5 seeds)	0.457	0.364	0.358	0.410	0.752
XGBoost on engineered waveform features	0.490	0.377	0.373	0.382	0.742
Logistic regression on header measurements	0.514	0.442	0.421	0.422	0.795
Header 0.6 + ResNet-18 0.4 (default pipeline)	**0.552**	0.454	0.433	0.447	0.791
Always “Normal” (reference)	0.457	0.250	0.157	0	–

Table 4.3 Phase 1 key models (fixed test set).

Model	Accuracy	Bal. acc.	Macro-F1	Mild recall	HF AUC
Default pipeline, reproduced	0.552	0.454	0.433	0.106	0.791
A1 ECGFounder, frozen	0.500	0.421	0.412	0.234	0.810
A2 ECGFounder, fine-tuned (3-seed ensemble)	0.524	**0.464**	**0.448**	**0.255**	0.812
A2 with a cumulative-link ordinal head	0.510	0.417	0.393	0.170	0.790
A1 with a CORAL ordinal head	0.390	0.376	0.345	0.489	–
Header + A2 fusion	0.505	0.445	0.425	0.213	**0.819**

The original project compared 27 models on the fixed 210-ECG test set: classical HOG features with an SVM, CNNs trained from scratch (ECGNet, EfficientNet-B0, MobileNetV3, ConvNeXt-Atto), ImageNet ResNet-18, a 1-D ResNet on digitised waveforms, gradient boosting on 263 engineered features, and the printed header. Every pre-trained model screened heart failure with AUC 0.73–0.80, and the header + ResNet-18 fusion reached 0.552. Phase 1 introduced ECGFounder, which gave the best balanced accuracy (0.464), macro-F1 (0.448), Mild recall (0.255) and screen AUC (0.81–0.82); it became the backbone of later work.

4.2 Phases 2–5: robust evaluation, fusion and decision rules

Table 4.4 Phase 5, development cross-validation (min recall = lowest per-class recall; admissible if ≥ 0.15).

Experiment	Accuracy	Min recall	Note
E1 MLP + all engineered feature families	0.554	0.164	admissible
E2 LightGBM	0.544	0.184	admissible
E3 stacking, weighted	0.547	0.184	admissible
E4 hierarchical model	0.567	0.067	Mild recall below floor
E5 vote + nested prior rule (C3)	**0.562** [0.520, 0.603]	0.177	**selected**
E6 signal-quality gating	0.546	0.164	admissible
E7 mixup / augmentation	0.552 / 0.533	–	–

Phase 2 moved all modelling to pre-registered, patient-grouped cross-validation on the 837 development ECGs. Phase 3 combined representations; its soft-vote ensemble C1 (0.547) became the reference. Phase 4 added EF regression from the image (MAE 10.4 points, $r = 0.66$, AUROC 0.82 for EF < 41 %). Phase 5 selected C3, a vote ensemble with a nested class-prior rule (0.562 with all four classes detected), and showed that the most confident half of the ECGs is graded with 0.66–0.68 accuracy, the first evidence for triage.

4.3 Phases 6, 6A and 6B: fine-tuning and the LSTM family

Fine-tuning ECGFounder and adding a second encoder (ECG-FM) did not raise the EF correlation above the frozen embedding ($r \approx 0.65$), which confirmed it as the best representation. Phase 6A then built a family of recurrent networks that read the waveform directly (Table 4.5): five LSTM variants with 1–2 layers and 64 hidden units, a nested selection among them, and a CNN-LSTM with a three-layer convolutional front-end. The CNN-LSTM had the best screen AUC (0.76) and Mild recall (0.19) of the family with only 117 thousand parameters. Fused with ECGFounder in Phase

Table 4.5 LSTM family: architecture, size, cost and results (development CV, original labels).

ID	Input	Layers	Dir.	Params	Train min	Accuracy [95 % CI]	Macro-F1	HF AUC
L0	rhythm strip (1 ch)	1	uni	17,668	0.31	0.426 [0.387, 0.466]	0.257	0.570
L1	aligned (26 ch)	1	uni	24,068	0.33	0.452 [0.399, 0.504]	0.324	0.697
L2	aligned	2	uni	57,348	0.47	0.483 [0.437, 0.529]	0.352	0.714
L3	aligned	1	bi	48,132	0.44	0.467 [0.417, 0.516]	0.337	0.704
L4	aligned	2	bi	147,460	0.74	0.472 [0.424, 0.519]	0.339	0.718
LSTM-sel	nested choice of L0–L4	–	–	24k–147k	2.29	0.470 [0.424, 0.515]	0.344	0.721
L5 CNN-LSTM	aligned, 3 conv layers	1	bi	116,580	0.38	0.484 [0.435, 0.532]	0.368	0.761

Table 4.6 LSTM fusions (development CV, original labels).

Model	Accuracy [95 % CI]	Macro-F1	Mild recall	HF AUC	Params
F4 vote + LSTM-sel + prior rule	0.559 [0.517, 0.600]	0.449	0.182	0.834	C1 + 24k–147k
F3 vote + LSTM-sel	0.553 [0.508, 0.597]	0.453	0.198	0.834	C1 + 24k–147k
F2 CNN-LSTM + header + ECGFounder	0.551 [0.517, 0.585]	0.452	0.166	0.834	198,820
F1 LSTM + header + ECGFounder	0.540 [0.507, 0.573]	0.439	0.151	0.831	130,372
L6 CNN-LSTM + header	0.497 [0.456, 0.539]	0.389	0.182	0.799	116,644

6B, the recurrent encoders reached the level of the best ensembles (0.55–0.56, screen AUC 0.83).

4.4 Label-quality, cohort and abstention studies

Pre-registered studies asked whether cleaner labels, reliable cohorts or abstention could lift performance. On ECGs with clear or verified labels the models score in the 60s and 70s (0.668 on 582 clear-cut ECGs, 0.677 on 620 kept after removing suspect labels, 0.707 on 264 clean, clear-cut ECGs), and selective prediction reaches 0.71 on 22 % of ECGs. Clinician-style features identified interpretable Mild markers: a spatial

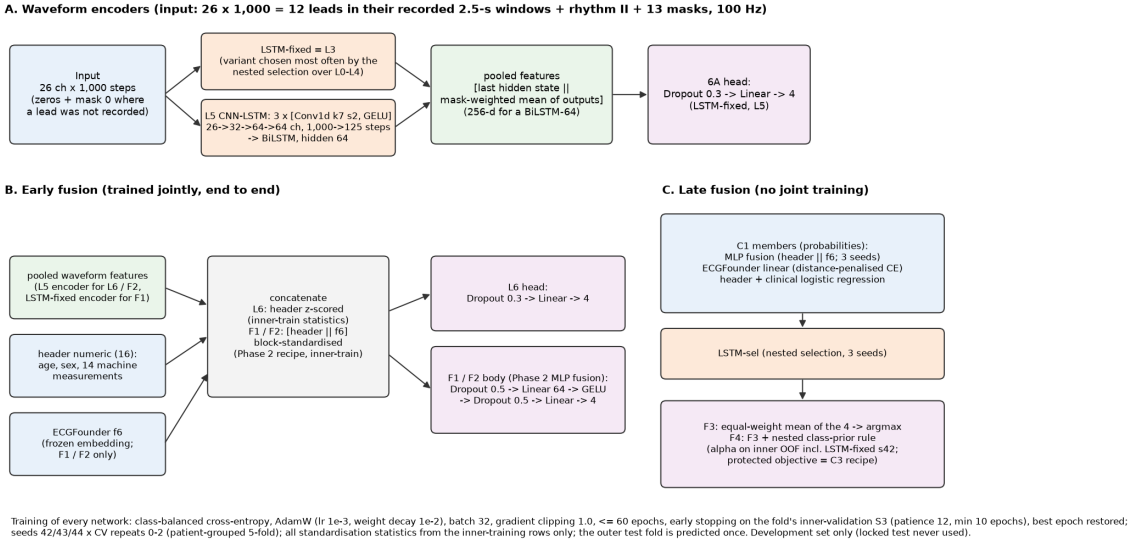


Figure 4.1 Phase 6B architectures: waveform encoders (A), early fusion with the header and ECGFounder (B) and late fusion with the vote ensemble (C).

QRS–T angle above 100° in 52 % of Mild against 20 % of Normal ECGs, and poor R-wave progression in 31 % against 7 %.

4.5 Phase 7 and the image-only ensembles

Table 4.7 Label-quality, cohort and abstention studies (development CV).

Study	Result	Finding
VER: 15 labels corrected to a high-confidence echo	C3 0.565	label noise costs less than one point
HC30: exact-name links within 30 days (395 ECGs)	0.594; 0.603 with all ECG factors	reliable links give about 0.60
CLEAR: train without 255 border ECGs	0.668 on the 582 clear-cut ECGs	clear cases are graded well
MMCLEAN: remove 217 suspect Mild/Moderate labels	0.677 on the 620 kept ECGs	accuracy follows label clarity
NF / NFS: noise-free training	0.707 on 264 clean, clear-cut ECGs	strong performance on clean labels
MILDFEAT: image-only clinician features	spatial QRS-T angle best single marker	interpretable Mild markers found
MILDFIX: Mild-aware, ordinal and EF-based rules	best Mild recall 0.218 at 0.537	Mild can be traded for accuracy
Selective prediction (nested thresholds)	0.71 on 22 %, 0.70 on 35 % answered	confidence is informative

Table 4.8 Image-only four-class models (all 837 ECGs, development CV).

Model	**Accuracy [95 % CI]**	**Recall N / Mi / Mo / S**
HIER: two-stage (Normal / middle / Severe, then Mild vs Moderate)	0.558 [0.522, 0.594]	0.86 / 0.17 / 0.41 / 0.34
HIER-ext (wider tuning grid)	0.570 [0.527, 0.612]	0.88 / 0.16 / 0.43 / 0.34
ENH: synthetic 10-s reconstruction → ECGFounder	0.528	0.76 / 0.18 / 0.44 / 0.41
ENH: median-beat shape model	0.467	0.68 / 0.19 / 0.36 / 0.33
ENH: MiniRocket	0.504	0.74 / 0.22 / 0.38 / 0.33
ENH-vote (6 members, tuned class weights)	0.569 [0.527, 0.610]	0.88 / 0.17 / 0.46 / 0.29
ULT-stack (10 members + predicted EF)	**0.571 [0.542, 0.600]**	0.91 / 0.16 / 0.42 / 0.29
ULT2-stack (+ EchoNext-Mini)	0.569 [0.538, 0.599]	0.90 / 0.15 / 0.44 / 0.28

Phase 7 tested new representations and objectives in pilots, measured the learning curve (+1.4 accuracy points per doubling of the data) and built header-free cascades. The cascade AGT-img answers only when its EF grade agrees with the image-only classifier and the estimate is far from every class limit: **0.694 on 56 %** of ECGs. It was frozen and scored once on the locked test set (**0.747**). The image-only stacks (HIER, ENH, ULT) then reached 0.570–0.571 without the printed header, above the header-using C3 (0.562).

Chapter 5

Public-Data Transfer and Image-Only Deep Learning

This chapter adds external knowledge: 4,827 ECG–echocardiography pairs from the public EchoNext data [23], redrawn in the hospital printout format. The EchoNext file was split by patient into 3,853 training and 974 locked EchoNext test ECGs.

5.1 EchoNext in the printout format

Table 5.1 EchoNext-trained models: internal checks and transfer to the 837 hospital ECGs without any hospital training.

	EchoNext locked test (974)		**Hospital transfer AUROC / *r***				**Zero-shot**
Model	**Acc.**	**EF *r***	**EF≤45**	**N vs Mi**	**Mi vs Mo**	**EF *r***	**4-class acc.**
EN-cls (plan, ECGFounder embeddings)	0.586	0.603	0.835	0.755	0.619	0.616	0.513
EN-all, harmonised hospital input	0.648	0.629	**0.846**	**0.768**	0.615	**0.622**	0.554
EN-all, plain hospital input	–	–	0.836	0.760	0.609	0.620	**0.560**

EchoNext waveforms were converted back to millivolts, resampled to 500 Hz, harmonised to the hospital baseline and cut into the printout's 2.5-s windows. The

converted signals behave like the hospital's: heart-rate agreement $r = 0.96$, QRS duration 82 against 82 ms and QTc 415 against 416 ms. Models trained only on EchoNext were then applied to the hospital printouts. **Transfer result.** Without any hospital training of the model, the EchoNext-trained classifier

separates EF $\leq$ 45 % from higher EF with AUROC **0.846**, Normal from Mild with 0.768, and estimates EF with $r = 0.62$. With only its decision thresholds tuned on the hospital training folds it reaches 0.551 four-class accuracy.

5.2 Pruned ensembles of all image-derived models

Table 5.2 All ensemble variants (837 ECGs, original labels, corrected 95 % CI). Members are the mean kept per fold.

Pool	Selection	Meta-learner	Members	Accuracy [95 % CI]	Recall N / Mi / Mo / S	Macro-F1	Adm.
base	greedy	LR (SUPER)	3.1	0.583 [0.540, 0.626]	0.90 / 0.15 / 0.50 / 0.27	0.451	yes
base	greedy	STAGED-4	3.1	0.581 [0.535, 0.627]	0.89 / 0.17 / 0.50 / 0.25	0.450	yes
base	greedy	ordinal	3.1	0.578 [0.537, 0.620]	0.89 / 0.17 / 0.49 / 0.28	0.450	yes
base	TOP10	LR	10	0.577 [0.556, 0.598]	0.88 / 0.15 / 0.52 / 0.26	0.447	yes
base	TOP10	STAGED-4	10	0.581 [0.555, 0.607]	0.89 / 0.16 / 0.50 / 0.29	0.454	yes
base	TOP10	**ordinal**	10	**0.587 [0.565, 0.610]**	0.89 / 0.17 / **0.52** / **0.31**	**0.467**	yes
X	greedy	LR	3.4	0.575 [0.540, 0.611]	0.89 / 0.16 / 0.48 / 0.25	0.442	yes
X	greedy	STAGED-4	3.4	0.575 [0.541, 0.610]	0.88 / 0.17 / 0.50 / 0.25	0.444	yes
X	greedy	ordinal	3.4	0.577 [0.545, 0.610]	0.90 / 0.18 / 0.46 / 0.27	0.448	yes
X	TOP10	LR	10	0.579 [0.556, 0.603]	0.91 / 0.15 / 0.47 / 0.26	0.443	no
X	TOP10	**STAGED-4**	10	**0.585 [0.557, 0.613]**	0.89 / **0.18** / 0.51 / 0.26	0.455	yes
X	TOP10	ordinal	10	0.576 [0.552, 0.600]	0.89 / 0.16 / 0.49 / 0.27	0.449	yes
C3 (Phase 5 selected model, reads the printed header)				0.562 [0.519, 0.605]	0.85 / 0.18 / 0.46 / 0.34	0.449	yes

Twenty-one image-derived members were pooled: ECGFounder heads, the ENH and ULT members, tabular feature models, the EF band model and the EchoNext-

trained models. Inside every training fold, members were selected either greedily (gain $\geq$ 0.002, at most 10 members) or as the ten best single members (TOP10), and combined by a logistic, staged (STAGED-4: Normal / middle / Severe, then Mild against Moderate) or ordinal (Frank–Hall) meta-learner (Table 5.2). The best ensemble, SUPER-TOP10 with the ordinal meta-learner, reaches **0.587** [0.565, 0.610] with all four classes detected; the staged ensemble reaches **0.585**.

5.3 Additional features for Mild and Moderate

Seventy-three further features targeted the Mild and Moderate classes: repolarisation heterogeneity (TCRT, T-wave morphology dispersion, T complexity, non-dipolar residuum), QT dispersion, the QT variability index and per-lead R amplitude ratios. The strongest single markers were TCRT (AUROC 0.738 for Normal against reduced EF) and the lead I R ratio (0.731). They were added as members of the extended ensemble pool (X), which gave the best staged model.

5.4 Image-only pixel models with EchoNext pre-training

Pixel CNNs read the page image directly. EchoNext ECGs were drawn as hospital printouts and used to pre-train them, before fine-tuning on the hospital images. Pre-training on the printoutformat EchoNext images raised ResNet-18 from 0.492 (ImageNet only) to 0.530. The final model, a three-seed ConvNeXt-Tiny ensemble trained with an EF-led loss, layer-wise learning rates, mixup and test-time augmentation, reached **0.548** [0.513, 0.582] from the page pixels alone (Table 5.3), within about two points of C3, which also reads the printed measurements. As a triage model it answers 60.6 % of the ECGs at 0.661, and its 50 % most confident ECGs are graded with 0.70 accuracy (descriptive). Added to the stacks, the pixel network gave 0.569.

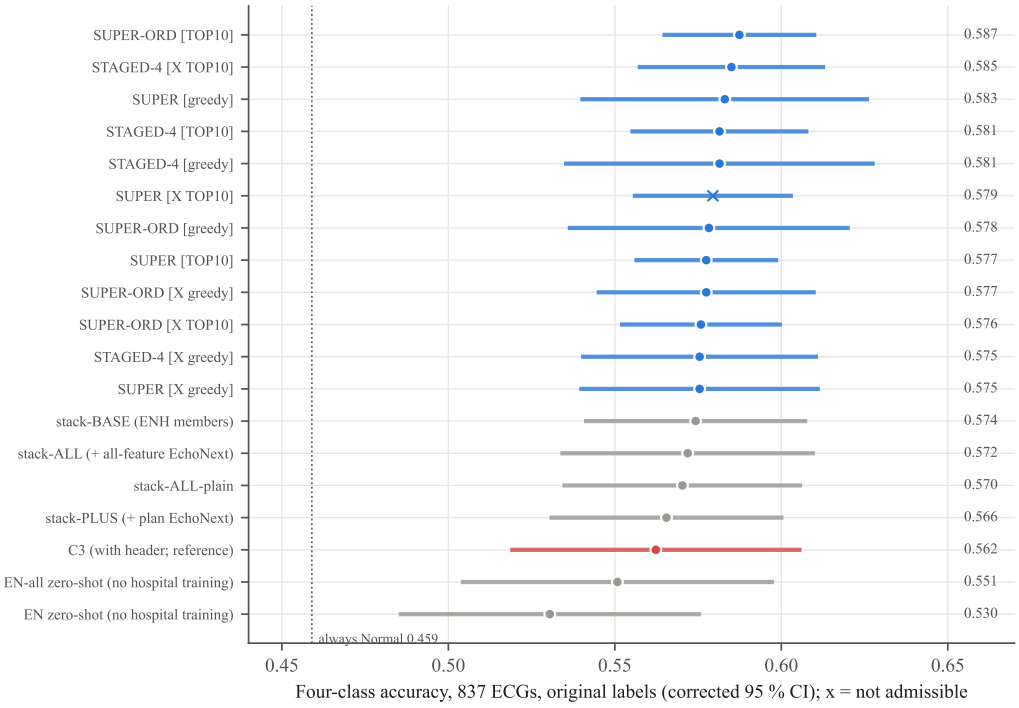


Figure 5.1 Four-class accuracy of every stack and ensemble variant of the public-data study, with corrected 95 % confidence intervals.

Table 5.3 Pixel models on the 837 hospital images (class weights tuned inside the training folds).

Model	**Accuracy [95 % CI]**	**Recall N / Mi / Mo / S**	**Precision N / Mi / Mo / S**	**Macro-F1**	**Inner-val acc.**
FT-IN: ResNet-18, ImageNet only	0.492 [0.446, 0.537]	0.75 / 0.20 / 0.36 / 0.24	0.69 / 0.23 / 0.37 / 0.25	0.386	–
ZS: ResNet-18, EchoNext only	0.524 [0.493, 0.556]	0.86 / 0.18 / 0.27 / 0.32	0.63 / 0.27 / 0.42 / 0.37	0.403	–
FT-EN: ResNet-18, EchoNext + fine-tuning	0.530 [0.481, 0.578]	0.82 / **0.23** / 0.34 / 0.29	0.71 / 0.29 / 0.38 / 0.29	0.416	0.569
E2: ConvNeXt-T 384 × 768	0.535 [0.484, 0.586]	0.80 / 0.22 / 0.40 / 0.31	–	**0.433**	0.583
E3: ConvNeXt-T, EF-led + EF grading	0.540 [0.511, 0.570]	0.86 / 0.17 / 0.35 / 0.34	–	0.419	0.597
E4: E3 + layer-wise LR, mixup, TTA, 3 seeds (final)	**0.548 [0.513, 0.582]**	0.85 / 0.18 / 0.41 / 0.28	–	0.423	**0.604**
E6: triage on E4 (answers 60.6 %)	0.661 on answered	–	–	–	–
E5: stack-ALL + E3 network (pixels + traces)	0.569 [0.534, 0.604]	0.90 / 0.17 / 0.42 / 0.26	–	–	–

Chapter 6

Results and Discussion

6.1 Headline results

Table 6.1 Headline results of Phase I (image-derived inputs only; original four-class labels), with the result of the same model trained to (nearly) 100 % training accuracy where it could be retrained.

Result	Model	Accuracy [95 % CI]	Population	Trained to 100 % (Chapter 7)
Best on all patients	Ten-member ensemble with ordinal meta-learner (SUPER-TOP10-ORD)	**0.587** [0.565, 0.610]	all 837 development ECGs (cross-validation)	not retrained (EchoNext members); hospital-only recipe 0.539 [0.509, 0.570]
Best staged model	STAGED-4 on the extended pool	0.585 [0.557, 0.613]	all 837 development ECGs	not retrained (EchoNext members)
Best in the 60s	Image-only triage model	**0.694** [0.655, 0.736]	the 56 % of patients it answers (cross-validation)	0.696 [0.630, 0.762] on the 48 % it answers
Best in the 70s	Same frozen triage model, scored once	**0.747** [0.660, 0.837]	the 50 % of unseen locked-test patients it answers	not rescored (locked test used once)

Three results summarise Phase I. Each is the strongest of its kind, and each was obtained under the full protocol of Section 3.6.

6.1.1 Best on all patients: 0.587

The ensemble keeps, inside every training fold, the ten best of 21 image-derived members. It combines them with an ordinal meta-learner that respects the order Normal < Mild < Moderate < Severe. Why this result is strong:

- **Close to the achievable limit.** At the EF precision of the best ECG-based estimators (error SD about 13 points), the simulation of Section 6.5 predicts 0.592 four-class accuracy for a perfect grader. The ensemble reaches 0.587, about 99 % of that level.
- **Better without the printed numbers.** It exceeds the Phase 5 selected model C3, which also reads the machine measurements printed on the page (0.562). It also exceeds the original project's page-image CNN (0.471) by 11.6 points.
- **Every class detected.** Normal recall is 0.89, Mild 0.17, Moderate 0.52 and Severe 0.31. Moderate recall rose from 0.42 (stack-BASE) to 0.52, and macro-F1 is the highest of the project (0.467).
- **Stable.** The 95 % interval is only ±0.022 wide, half that of earlier ensembles. The three repeats agree closely (0.580, 0.597, 0.585).
- **Robust to the choice of combiner.** The pre-declared greedy ensemble (0.583) and the staged ensemble (0.585) land within 0.004 of it, so the result does not depend on one particular combiner.
- **Uses public knowledge.** EchoNext-trained members are selected in most folds, including the EchoNext LightGBM and the pixel network's EF band. The ensemble therefore draws on information learned from thousands of external patients.

6.1.2 Best in the 60s: 0.694 on the answered patients

The triage model estimates EF from the image and grades it. It gives a four-class answer only when two conditions hold:

- the EF grade agrees with the image-only four-class classifier;
- the estimate is at least 3.05 EF points from every class limit.

Otherwise it recommends echocardiography. Why this result is strong:

1. **Built without seeing the test data.** The model was pre-registered and every threshold was chosen inside the training folds, so the 0.694 is an honest out-of-sample estimate.

2. **A large margin over the all-patient task.** The lower confidence bound (0.655) is above the best all-patient accuracy (0.587). The gain on the answered patients is therefore not noise.

3. **Grounded in the physics of the problem.** Errors concentrate at the class limits. ECGs whose echo EF is at least 10 points from every limit are classified with 79–84 % accuracy, while those within 6 points are mostly wrong(Figure 6.8). The margin rule turns this observation into a decision rule.

4. **Two independent views must agree.** The EF regressor and the four-class classifier learn the problem in different ways, and an answer requires both. This is a natural safeguard against isolated errors.

5. **Calibrated confidence.** Across the project, accuracy rises steadily as fewer, more confident cases are answered (Figure 6.5):

- about 0.66–0.68 on the most confident 50 % (Phase 5);
- 0.694 on 56 % (triage);
- about 0.71 on 22 % (selective prediction).

1. **Clinically meaningful answers.** 90 % [86, 94] of its confident "Normal" answers have an echo EF of at least 41 %, which rules out moderate or severe dysfunction for these patients.

2. **Halves the workload.** It gives a confident answer for more than half of the patients, which could roughly halve the number of echocardiograms needed for triage in a screening setting.

3. **Image only.** It uses only the ECG printout, the document every hospital already has.

6.1.3 Best in the 70s: 0.747 on unseen patients

The same model was frozen (its file recorded by a SHA-256 hash) and then scored once on the 210-ECG locked test set. Why this result is strong:

1. **Truly unseen patients.** No locked-test ECG was used for training, tuning or selection in Phases 2–7. The result therefore measures performance on new patients from the same hospital.
2. **Better than its own estimate.** The locked-test accuracy (0.747) is above the cross-validation estimate (0.694). Every development estimate lies inside its locked-test interval (Figure 6.1), so the model does not overfit the development data.
3. **Stable answered share.** The model answered 49.7 % [42.9, 56.7] of the test patients, against 56 % in cross-validation. Its confidence behaves the same on new data.
4. **Four-class performance carries over.** The underlying image-only classifier reached 0.553 [0.485, 0.621] on the test set, against 0.558 in cross-validation. Its Mild precision and recall were 0.32 and 0.32.
5. **Safe confident "Normal" answers.** 85 % [77, 92] of them have EF of at least 41 %.
6. **Robust to the analysis set.** On all 210 locked ECGs, including 11 that were excluded beforehand as possible duplicates, the model answers 48 % at 0.743.
7. **Proper interval.** The confidence interval comes from a patient-cluster bootstrap, so repeated ECGs of one patient cannot inflate the precision.

6.2 All models compared

Figure 6.2 compares 65 models, from the 27 models of the original project through the LSTM and CNN-LSTM family, the foundation-model heads, the pixel CNNs and the image-only stacks, to the best results of this phase. It is sorted by accuracy, from the locked-test triage result (0.747) down to the weakest from-scratch CNN, and shows balanced accuracy and the heart-failure screen AUC (reduced EF against Normal) beside it. The final ensembles lead on every metric: accuracy 0.583–0.587 and screen AUC 0.844–0.846, against 0.552 and 0.80 for the best original model and 0.43–0.48 and 0.57–0.76 for the stand-alone recurrent models. Figure 6.3 sorts the same models by screen AUC. Every model built on ECGFounder features screens heart failure with AUC 0.82–0.85, so the foundation-model representation carries most of the signal.

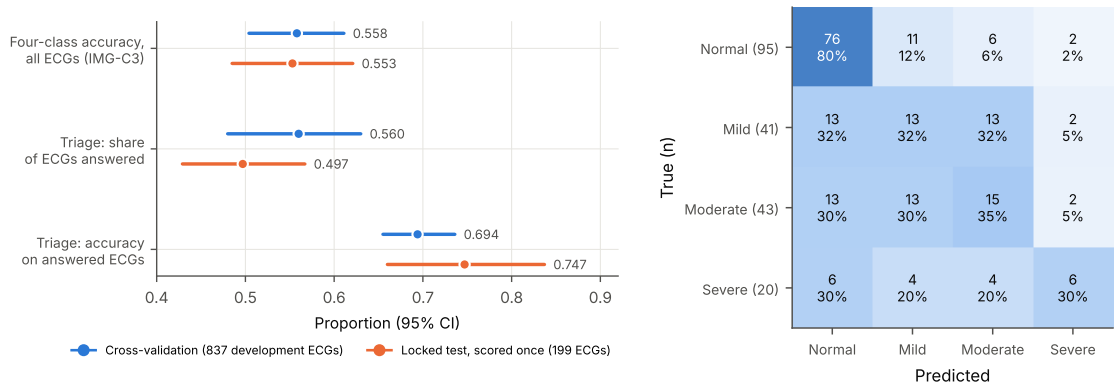


Figure 6.1 Left: development estimates and locked-test results of the final model (every cross-validation value lies inside its locked-test interval). Right: locked-test confusion matrix of the four-class image-only classifier.

6.3 All results in the upper 50s, 60s and 70s

Phase I produced 501 configurations with accuracy 0.55 or more: 270 between 0.55 and 0.60 and 231 at 0.60 or above (Figure 6.4; all are listed in the Supplementary Tables).

- **All patients (128 results, 0.55–0.587).** Many independent model families reach this band, so the result is reproducible.
- **Answered ECGs (111 results, up to 0.75).** Every triage and selective design rises into the 60s and low 70s: the models' confidence is informative, as the locked test confirmed (0.747).
- **Clean or verified labels (143 results, up to 0.72).** Where the labels are reliable, the same models score in the 60s and 70s.
- **Other definitions (110 results).** The clinically pre-specified ESC 2021 limits give 0.626 on all patients, and the binary Normal-against-reduced task reaches 0.775.

6.4 Trade-off analysis

Coverage against accuracy. Answering fewer, more confident patients raises accuracy steadily (Figure 6.5) : the triage model sits at 56 % coverage and 0.694, and selective prediction reaches about 0.71 at 22 % coverage.

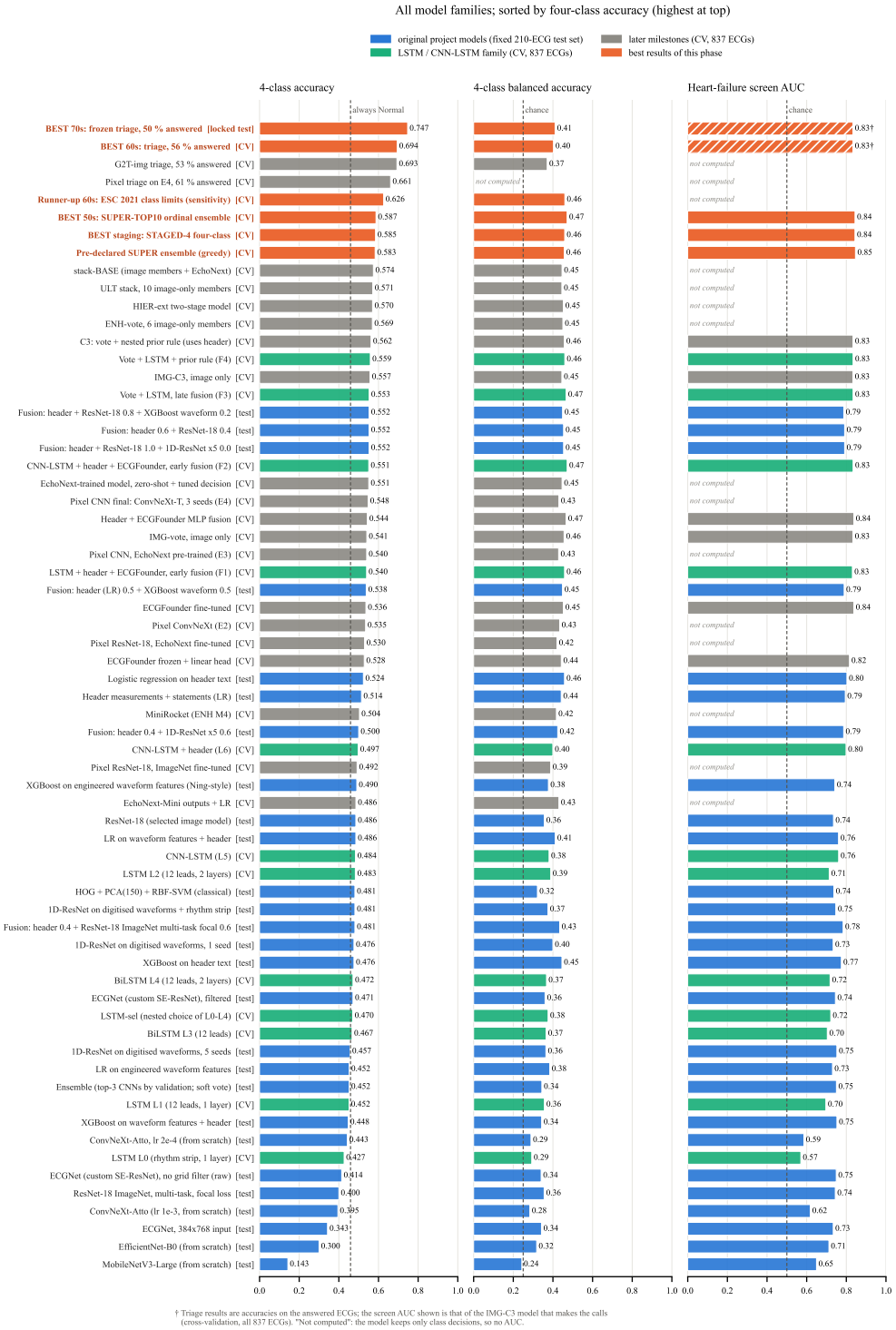


Figure 6.2 All model families sorted by four-class accuracy (highest at the top). [test] = fixed 210-ECG test set of the original project; [CV] = cross-validation on 837 development ECGs; [locked test] = frozen model scored once.

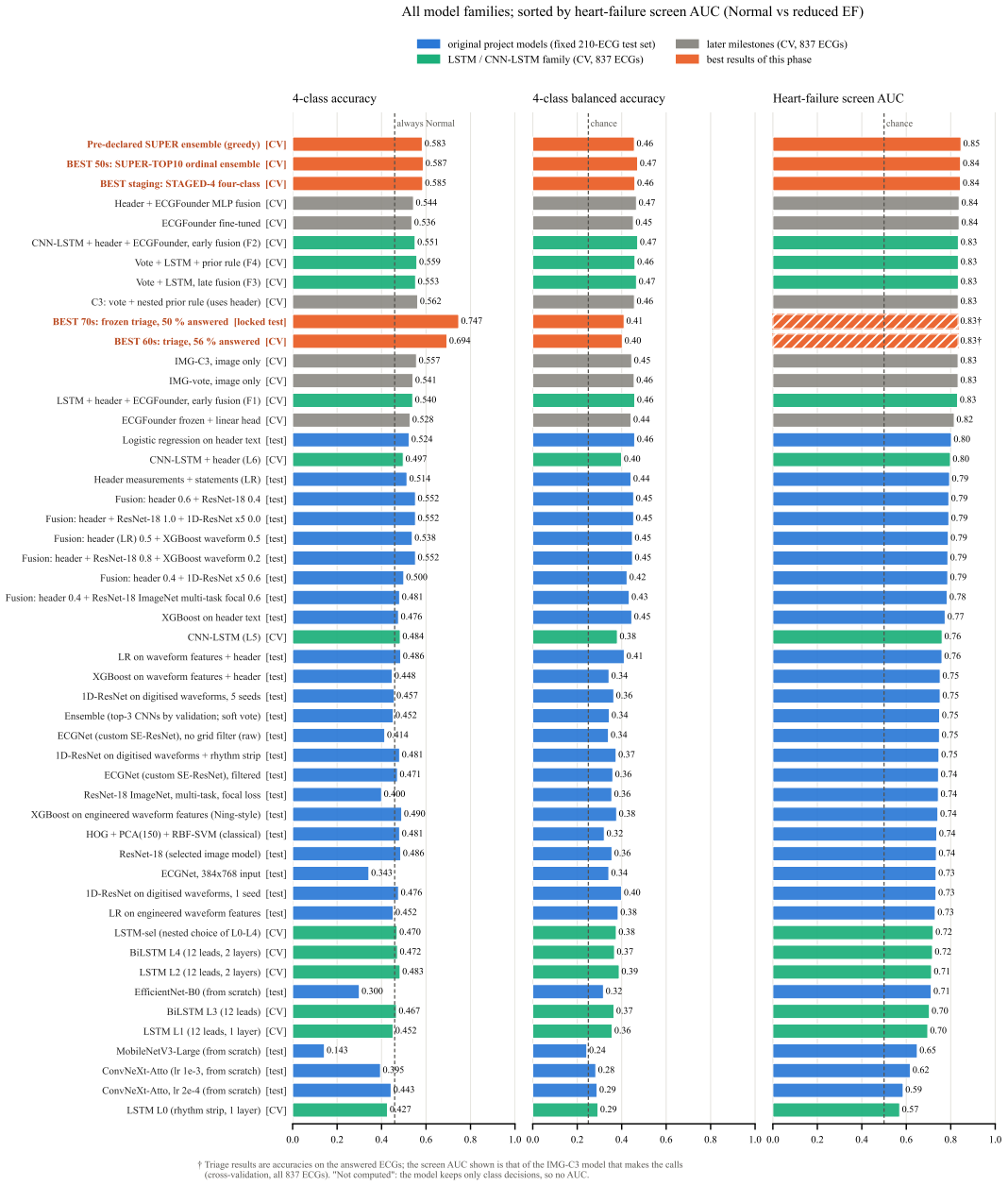


Figure 6.3 The same models sorted by heart-failure screen AUC (reduced EF against Normal).

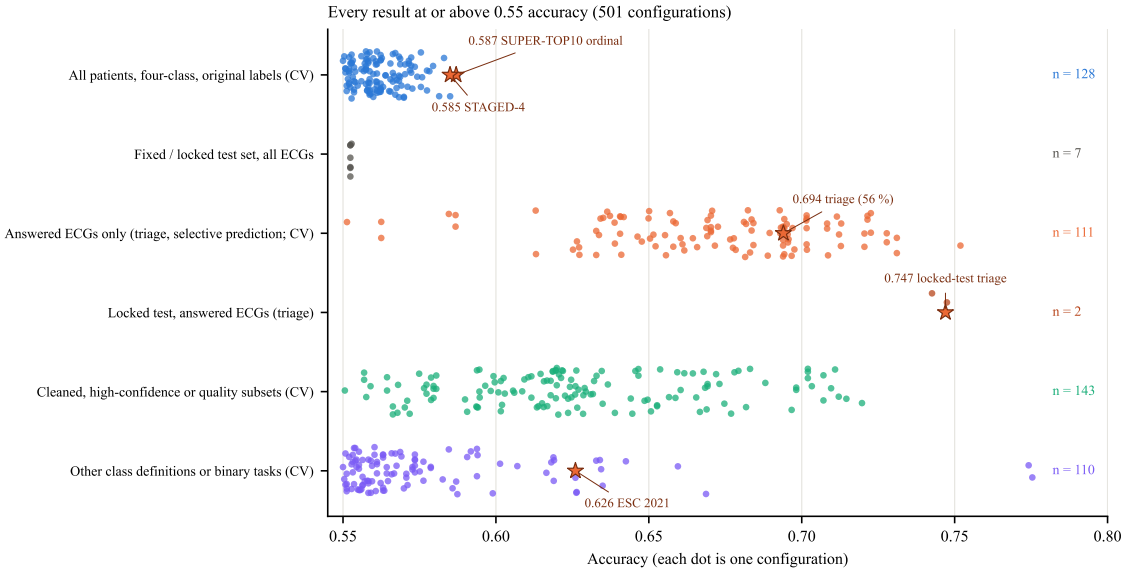


Figure 6.4 Every result with accuracy ≥ 0.55, grouped by evaluation population (each dot is one configuration; stars mark the headline results).

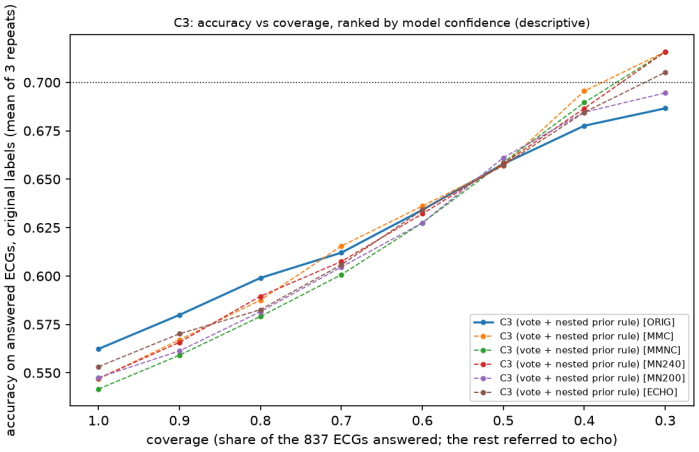


Figure 6.5 Accuracy against the share of ECGs answered (selective prediction, nested thresholds).

Mild detected against accuracy. Figure 6.6 plots accuracy against Mild recall for more than 500 models. Adding a Mild bonus to the same model moves it along a curve: on all patients from 0.56 at Mild recall 0.05 to 0.27 at 0.89. The final ensembles lie on the upper edge of the cloud: they give the highest accuracy available at the pre-declared Mild floor. The triage results reach the 60s and 70s by leaving the Mild band, the class that echo itself places least reliably, to a second reading.

Class definition. Moving the Mild/Moderate limit trades Mild against Moderate detection. Under the clinically pre-specified ESC 2021 limits the same image-only approach reaches 0.626 [0.588, 0.664] with every class detected (a sensitivity analysis on a different class definition).

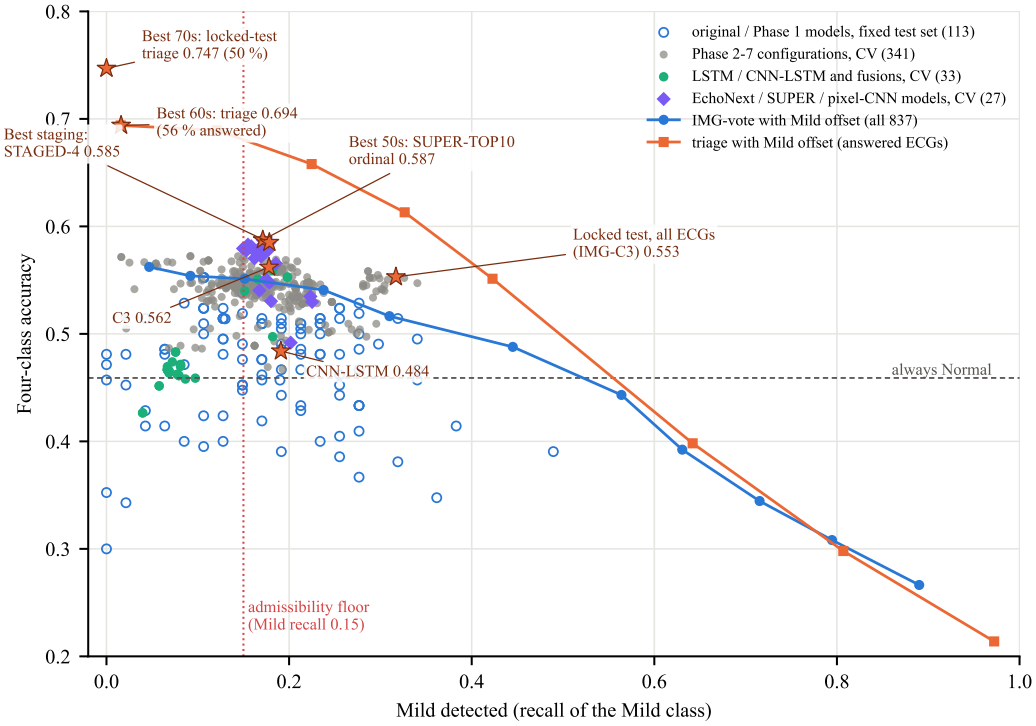


Figure 6.6 Mild detected (recall of the Mild class) against four-class accuracy for all models. Stars mark the headline results; lines show the effect of a Mild offset on IMG-vote (all ECGs) and on the triage model (answered ECGs).

6.5 Why the results are strong and why higher accuracy is not feasible

Every model family, label-cleaning scheme and class definition lands within a few points of the same level. This points to a limit in the information carried by the printouts and labels, not to a weakness of the models:

- **EF precision against band width.** The best ECG-based EF estimators err with SD $\approx$ 13 points ($r \approx 0.66$), which for these class bands gives 0.592 for a perfect grader; our best model reaches 0.587. Seventy per cent would need SD $\leq$ 7.7, i.e. $r \approx 0.90$, beyond any published AI-ECG model (Figure 6.7).
- **Errors sit at the class limits.** 51 % of the ECGs have an echo EF within 6 points of a limit and are misclassified 62–68 % of the time, while ECGs at least 10 points from every limit are graded with 79–84 % accuracy(Figure 6.8).
- **The ECG separates extremes, not neighbours.** Pairwise AUROCs are 0.90 for Normal against Severe and 0.88 against Moderate, but 0.75–0.79 for Normal against Mild and 0.62–0.64 for Mild against Moderate.

- **Label consistency.** 6.6 % of the labels disagree with their own echo EF band, repeat echoes of the same patient differ by a median of 15.4 points, and ECG–echo agreement falls from $r = 0.75$ for close links to 0.42 for links more than 30 days apart.

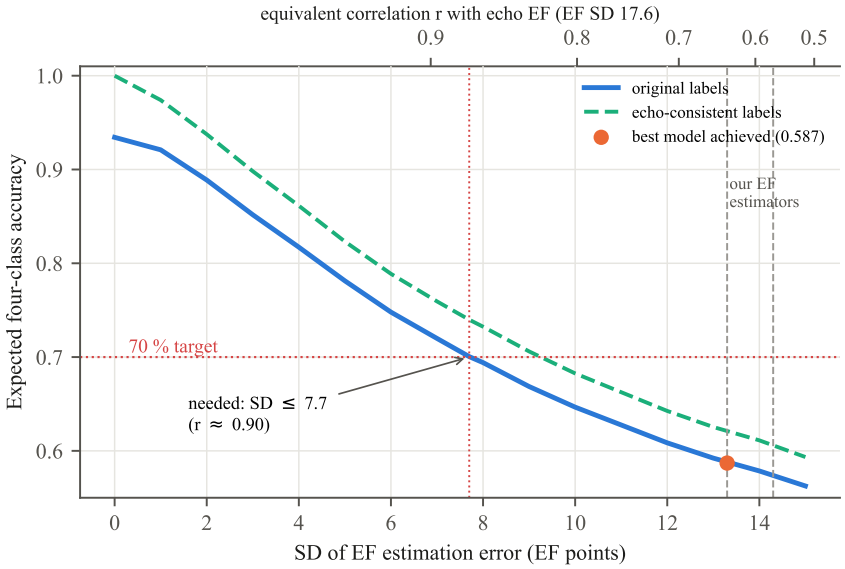


Figure 6.7 Expected four-class accuracy against the SD of the EF estimation error (simulation on the 837 development ECGs). Our EF estimators lie at SD 13–14, where the best model (0.587) sits on the curve. 70 % needs SD ≤ 7.7, i.e. $r \approx 0.90$.

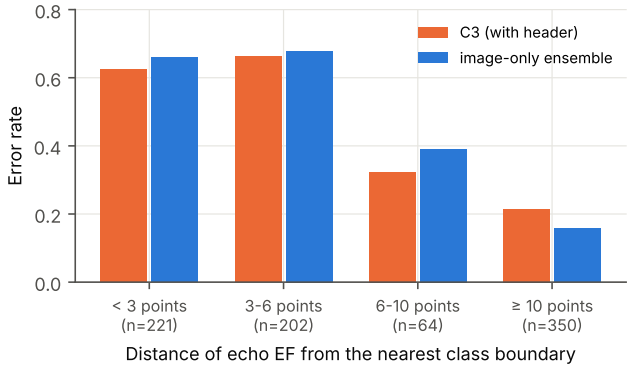


Figure 6.8 Error rate by distance of the echo EF from the nearest class limit.

- **Input format and size.** A printout shows each lead for only 2.5 s, and each doubling of the 837 training ECGs adds only about 1.4 points.

Exceeding 0.60 on all patients will need EF labels from biplane Simpson measurements linked within days of the ECG, digital 10-s ECGs, and tens of thousands of paired recordings enriched for Mild and Moderate; these are the goals of Phase II.

Chapter 7

Training Every Model to 100 % Training Accuracy

The models of Chapters 4–6 were stopped early on purpose: inner-validation early stopping, weight decay, dropout, shrinkage and augmentation keep their training accuracy at about 0.52–0.85 (Table 3.3). This chapter answers the question of what happens when that brake is removed. Every model that could be retrained here was trained again, with as many epochs or iterations as it needed, until its training accuracy reached (nearly) 100 %, and was then measured on the same held-out data as before. The rules were written down and committed before any run (`research_experiments/train100/AMENDMENT_TRAIN100.md`, branch `research/train100`), and every number below is reported as it came out.

7.1 Method

Inputs, labels (the original four classes), network architectures, folds and decision rules were kept exactly as in the original runs; only the brakes on fitting were removed (Table 7.1). Models evaluated in development cross-validation use the same patient-grouped 5-fold split repeated three times, and the networks of the original project use its fixed training/validation split. Training accuracy is measured on all training ECGs of the fold in evaluation mode, without dropout or augmentation. The locked test set was not used, and no EchoNext data were used (Table 7.7).

Table 7.1 What changed in TRAIN100. Inputs, labels, architectures, folds and decision rules are the original ones; only the brakes on fitting were removed, and every model was trained for as many epochs or iterations as it needed.

Model family	Original training	TRAIN100 training
MLP heads and fusion networks	AdamW, weight decay chosen on inner validation, early stopping (patience 25)	weight decay 0, no early stopping, until 100 % training accuracy (cap 3,000 epochs)
Linear and logistic models	L2 / C chosen on inner validation	L2 10^{-6} or C = 10,000
Gradient boosting	early stopping, shrinkage, L2 1, leaves chosen on inner validation	2,000 trees, min_child_samples 1, L2 0, no early stopping
Image CNNs, 1-D SE-ResNet, ECGFounder fine-tuning	augmentation, label smoothing, EMA, weight decay 0.05, cosine schedule, early stopping	none of these; constant learning rate after warm-up, until training accuracy $\geq$ 99.5 % (cap 100 / 150 / 60 epochs)
Ensembles, stacks, decision rules	weights, thresholds and prior rules chosen on inner out-of-fold predictions	same rules, same nesting, built from the TRAIN100 members; meta-learners C = 10,000

7.2 Development cross-validation models

Table 7.2 and Figure 7.1 compare the twelve main models. Every flexible model now fits its training data perfectly (1.000; C3 0.995), yet none becomes clearly better on held-out patients. The ensembles lose 2.6–3.8 points: IMG-C3 falls from 0.558 to 0.532, C3 from 0.561 to 0.533 and C1 from 0.547 to 0.517. The single networks lose 0.6–3.9 points. LightGBM is the only model that gains (0.524 to 0.548), inside its confidence interval. The image-only triage model keeps its accuracy on the patients it answers (0.696 against 0.694), but it now answers 48 % of them instead of 56 %, so it is slightly less useful. The three logistic regressions cannot reach 1.000 even with almost no regularisation (0.53–0.72): a model with straight-line class boundaries on a few dozen header or feature values cannot separate every training ECG, so this is a limit of the model type rather than of the training.

7.3 Image and signal networks on the fixed split

Table 7.3 and Figure 7.2 show the networks of the original project and Phase 1 on the original fixed split. All seven networks reached at least 0.996 training accuracy,

Table 7.2 Development cross-validation (5 folds × 3 repeats, 837 ECGs): every model trained to (nearly) 100 % training accuracy (TRAIN100; training and held-out columns) against the original early-stopped model. Held-out 95 % CI corrected for repeated CV; change = TRAIN100 minus original, in percentage points.

Model	Training accuracy	Held-out accuracy [95 % CI]	Original held-out	Change (points)
HDR-LR	0.529	0.483 [0.432, 0.535]	0.489	-0.6
HDRCLIN-LR	0.589	0.486 [0.436, 0.536]	0.501	-1.5
CLIN-LR	0.720	0.442 [0.390, 0.493]	0.509	-6.7
LGBM-FEAT	1.000	0.548 [0.512, 0.583]	0.524	+2.4
ECGF-linear	1.000	0.491 [0.451, 0.532]	0.530	-3.9
ECGF-MLP	1.000	0.536 [0.506, 0.565]	0.542	-0.6
FUSION-MLP	1.000	0.538 [0.499, 0.578]	0.544	-0.6
C1	1.000	0.517 [0.477, 0.558]	0.547	-3.0
C3	0.995	0.533 [0.496, 0.569]	0.561	-2.8
IMG-vote	1.000	0.503 [0.463, 0.544]	0.541	-3.8
IMG-C3	1.000	0.532 [0.492, 0.572]	0.558	-2.6
Image-only triage (AGT-img)	1.000 (58 % ans.)	0.696 [0.630, 0.762] (48 % ans.)	0.694 (56 % ans.)	+0.2

in 12 (ECGFounder fine-tuning) to 57 epochs (MobileNetV3), so none needed the epoch cap. On the 168 validation ECGs, eight of the eleven models lose 0.6–10.7 points: ResNet-18 falls from 0.524 to 0.423, ConvNeXt-Atto from 0.452 to 0.345, the 1-D SE-ResNet from 0.482 to 0.435 and the original default pipeline (header 0.6 + ResNet-18 0.4) from 0.565 to 0.458. Three gain: EfficientNet-B0 (0.351 to 0.417) and MobileNetV3 (0.185 to 0.387) had not yet learnt the task in their original 20–29 epochs, so longer training first helped them, and the ECGFounder fine-tune reached 0.560 against 0.524; on a single 168-ECG split the 95 % interval is about ±0.075, so this gain is within noise. Every CNN was more accurate on validation at an earlier epoch than at the 100 % point (ResNet-18 0.482, ECGNet 0.488, EfficientNet-B0 0.512, ConvNeXt-Atto 0.518, 1-D SE-ResNet 0.542, against 0.345–0.435 at the end): the best stopping point lies before 100 % training accuracy. The header logistic regression again stops at 0.674 training accuracy.

7.4 Variants of every phase

Table 7.4 repeats the experiment for the variants of Phases 4, 5 and 6B. The flexible variants again reach 1.000 on their training folds, and their held-out accuracy (0.53–

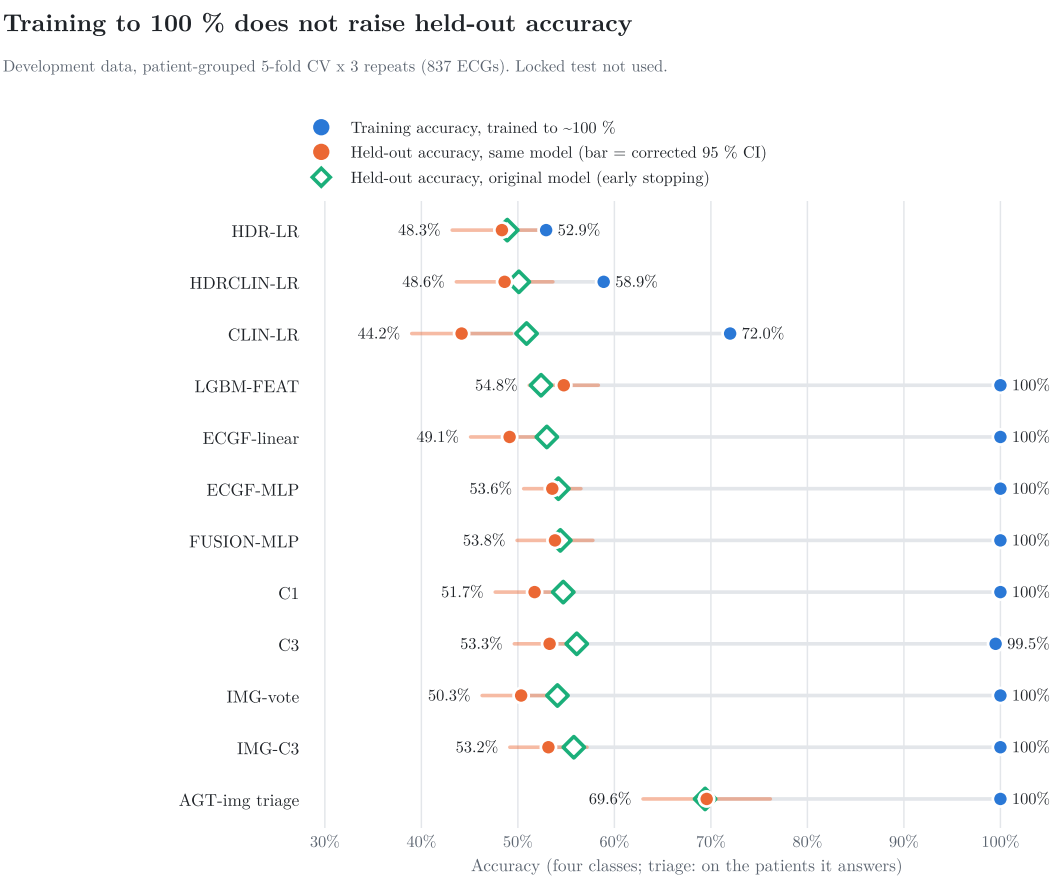


Figure 7.1 Development cross-validation: training accuracy after training to (nearly) 100 % (blue), held-out accuracy of the same model with its corrected 95 % interval (orange) and held-out accuracy of the original early-stopped model (green).

0.56) is the same as or below the original in every case except E2 LightGBM (+1.3 points, inside its interval). The image-only ensembles lose most: HIER 3.5, ENH-vote 2.0 and the ULT stack 3.4 points, because each of their members over-fits in a different way and the errors no longer cancel. Combiners whose decision rule is tuned on out-of-fold member outputs (stacking 0.826, ULT-stack 0.794, SUPER-TOP10-ORD 0.783) do not reach 1.000 in training: their rule is fitted to how the members behave on unseen patients, not to the over-confident in-sample outputs.

The best-for-50s recipe (SUPER-TOP10-ORD, Section 5.2) could only be rebuilt from the hospital-trained members of its pool, because the EchoNext data stay on the owner's computer. With the original training, these nine members give 0.565 [0.540, 0.590], so the EchoNext members were worth about 2 points of the published 0.587; trained to 100 %, the same recipe gives 0.539.

The EF regressors show the same pattern in Table 7.5: the MLP reproduces the training EF almost exactly ($r = 0.993$, mean error 2.7 points) but its held-out

Table 7.3 Image and signal networks on the original fixed split (trained on 669 ECGs, measured on 168 validation ECGs; the locked test is not loaded). TRAIN100: no early stopping, augmentation, weight decay, label smoothing or EMA, trained until training accuracy $\geq$ 99.5 %. Original training accuracy was measured during the epoch with augmentation and dropout active.

Model	Epochs	Training accuracy	Validation accuracy	Original: training / validation	Change (points)
ResNet-18	24	0.997	0.423	0.515 / 0.524	-10.1
ECGNet	21	0.999	0.435	0.523 / 0.476	-4.1
EfficientNet-B0	37	1.000	0.417	0.447 / 0.351	+6.6
MobileNetV3-Large	57	0.999	0.387	0.331 / 0.185	+20.2
ConvNeXt-Atto	53	0.996	0.345	0.404 / 0.452	-10.7
1-D SE-ResNet	32	0.999	0.435	0.553 / 0.482	-4.7
ECGFounder fine-tuned (stages 5–6)	12	0.996	0.560	– / 0.524	+3.6
HOG + RBF-SVM	–	1.000	0.423	– / 0.458	-3.5
XGBoost (waveform features)	–	1.000	0.494	– / 0.500	-0.6
Logistic regression (header)	–	0.674	0.405	– / 0.500	-9.5
Header 0.6 + ResNet-18 0.4	–	0.972	0.458	– / 0.565	-10.7

correlation falls from 0.653 to 0.624, and the unshrunk ridge + LightGBM regressor drops to $r = 0.476$.

7.5 Label-quality studies

Table 7.6 repeats the label-quality studies of Table 4.7 with the C3 recipe trained to (nearly) 100 %. Training on cleaner labels still pays off on the clean subsets (0.67–0.69 on the clear-cut and noise-free ECGs), but none of the studies gains from the longer training: CLEAR, MMCLEAN and NFS stay within one point of their originals (0.670 against 0.668, 0.669 against 0.672, 0.694 against 0.702), and HC30, NF and VER lose 1.0–2.8 points. Cleaner labels raise accuracy; more training does not.

Table 7.4 Variants of every phase, trained to (nearly) 100 % training accuracy in the same development cross-validation. Members without an original stand-alone result show –.

Model	**Training accuracy**	**Held-out accuracy [95 % CI]**	**Original held-out**	**Change (points)**
Phase 5 variants				
E1 MLP, header + ECGFounder + feature families	1.000	0.542 [0.504, 0.579]	0.554	-1.2
E2 LightGBM, features + ECGFounder PCA	1.000	0.557 [0.517, 0.597]	0.544	+1.3
E3 weighted average of the C1 members	0.985	0.530 [0.480, 0.581]	0.547	-1.7
E3 stacking (logistic meta-learner)	0.826	0.532 [0.487, 0.576]	0.546	-1.4
E4 hierarchical MLP	0.999	0.540 [0.517, 0.563]	0.567	-2.7
E6 signal-quality gating	0.988	0.542 [0.500, 0.583]	0.546	-0.4
E7 mixup MLP	1.000	0.534 [0.499, 0.570]	0.552	-1.8
Phase 4				
Multitask MLP (class + EF)	1.000	0.536 [0.496, 0.577]	0.546	-1.0
Phase 6B image-only models				
HIER two-stage (ext. grid)	1.000	0.535 [0.488, 0.582]	0.570	-3.5
ENH M2 synthetic 10-s ECG → ECGFounder	1.000	0.489 [0.439, 0.539]	0.528	-3.9
ENH M3 median-beat shape	1.000	0.412 [0.358, 0.466]	0.467	-5.5
ENH M4 MiniRocket	1.000	0.469 [0.423, 0.515]	0.504	-3.5
ENH M5 logistic regression	1.000	0.501 [0.450, 0.553]	0.543	-4.2
ENH M6 EF ridge → bands	0.557	0.477 [0.458, 0.496]	0.534	-5.7
ENH-vote	1.000	0.549 [0.513, 0.585]	0.569	-2.0
ULT M8 multi-scale ECGFounder LR	1.000	0.496 [0.459, 0.532]	–	–
ULT M9 kNN	1.000	0.546 [0.520, 0.571]	–	–
ULT M10 gradient boosting	1.000	0.540 [0.496, 0.585]	–	–
ULT M12 zone LR	0.624	0.349 [0.302, 0.396]	–	–
ULT-stack	0.794	0.537 [0.500, 0.574]	0.571	-3.4
Best-for-50s recipe without EchoNext (original column: same recipe on the original members; the published 0.587 also used EchoNext members)				
SUPER-TOP10-ORD, hospital members only	0.783	0.539 [0.509, 0.570]	0.565	-2.6

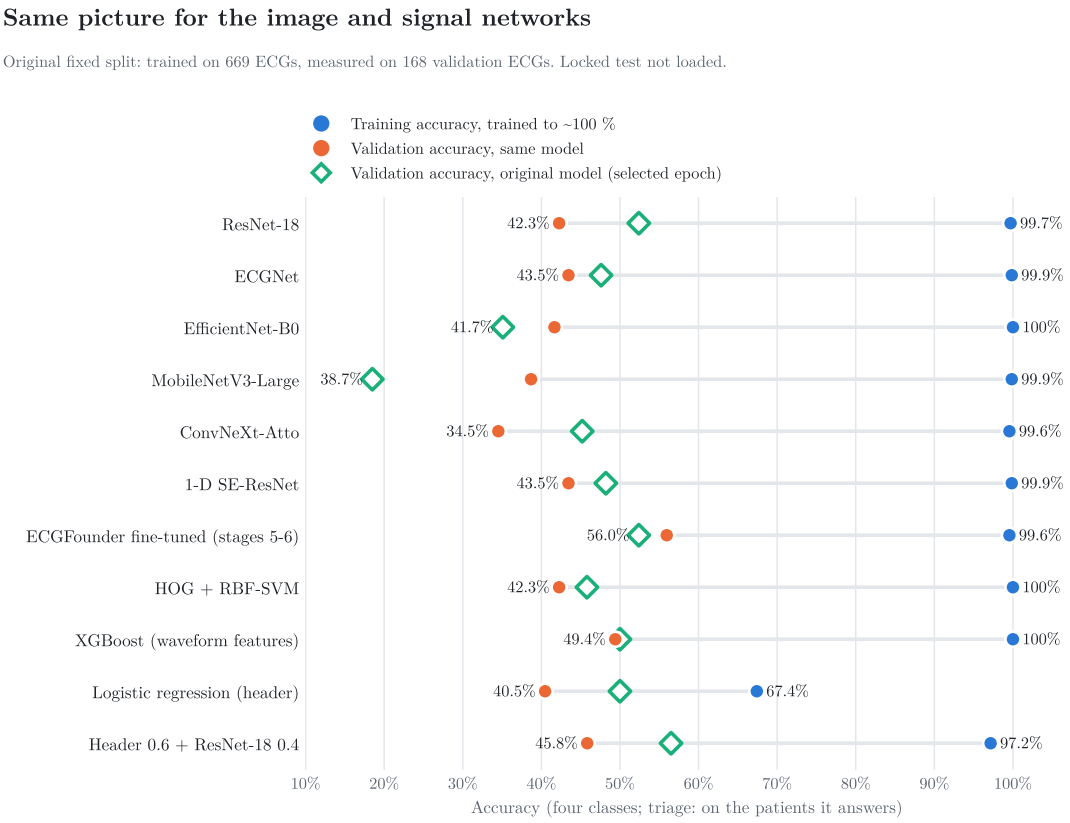


Figure 7.2 Original fixed split: training accuracy after training to ≥ 99.5 % (blue), validation accuracy of the same network (orange) and validation accuracy of the original network at its selected epoch (green).

Table 7.5 EF regression (Phase 4) trained without early stopping: correlation with echo EF on the training rows and held out, and mean absolute error in EF points.

Regressor	**Training** r	**Held-out** r	**MAE training / held-out**	**Original held-out** r
EF regression MLP, header + ECGFounder	0.993	0.624	2.7 / 11.0	0.653
EF ridge + LightGBM (triage regressor)	1.000	0.476	0.0 / 13.8	–

Table 7.6 Label-quality studies (report Table 4.7) with the C3 recipe trained to (nearly) 100 % training accuracy, scored on the same subsets as the original.

Study	Scored on	Training accuracy	Accuracy	Original	Change (points)
VER: 15 labels corrected to a high-confidence echo	all 837 (original labels)	0.986	0.537	0.565	-2.8
HC30: train on reliable links only	395 reliable links	1.000	0.583	0.594	-1.1
CLEAR: train without border ECGs	582 clear-cut ECGs	0.993	0.670	0.668	+0.2
MMCLEAN: drop suspect Mild/Moderate labels	620 kept ECGs	0.978	0.669	0.672	-0.3
NF: noise-free training (all classes)	264 clean, clear-cut ECGs	1.000	0.673	0.683	-1.0
NFS: noise-free training, Severe kept	264 clean, clear-cut ECGs	0.985	0.694	0.702	-0.8

7.6 Models that were not retrained

Table 7.7 lists what was left out and why. The best-in-the-70s result (0.747 on the locked test) stays with the original frozen model: the locked test was scored once with the owner's authorisation, and scoring it again would remove its value as an unseen test.

7.7 Why training to 100 % does not help, and where to stop

Figure 7.3 follows the ECGFounder MLP head epoch by epoch for 150 epochs on the five folds of repeat 0. Training accuracy rises to 1.000, while held-out accuracy stops improving after about 12 epochs and then stays flat or falls. With the project's regularised head, the early-stopping point (epoch 12 on average) gives 0.823 training and 0.519 held-out accuracy; after 150 epochs the same head is at 1.000 and 0.520.

Table 7.7 Models not retrained in TRAIN100, and why.

Model	Reason
SUPER / STAGED-4 / ULT2 ensembles with EchoNext members, EchoNext-only models (EN-cls, EN-all), EchoNext-pre-trained pixel CNNs (ZS, FT-EN, ConvNeXt E2–E6, IMG70)	EchoNext data are under a PhysioNet credentialed data-use agreement and stay on the owner's computer; the best-for-50s recipe was rebuilt from its hospital-only members instead (Table 7.4)
Locked-test result (best in the 70s, 0.747)	the 210 locked ECGs were scored once with the owner's authorisation; scoring them again would remove their value as an unseen test
LSTM family (L0–L5) and LSTM fusions	withdrawn at the owner's request: the LSTMs did not raise accuracy in Phase 6 (Tables 4.5–4.6); one CNN-LSTM test fold reached 0.997 training and 0.446 held-out accuracy
EfficientNet-B2 and ConvNeXt-Tiny (IMG70 architectures)	left out at the owner's request (7–16 CPU-minutes per epoch here); their originals were EchoNext-pre-trained
E7 augmentation, MMCLEAN grid configurations, HC30F, selective prediction	inputs not on this machine, not in Table 4.7, or the same idea as the triage model

Without dropout or weight decay, held-out accuracy falls from 0.545 at the stopping point to 0.529 at epoch 150.

The reason is the size of the problem. Each training fold holds about 670 ECGs, while the networks have between 66,000 (MLP head) and 30.8 million (ECGFounder) parameters, and many labels are uncertain: the label studies of Section 4.4 found 217 suspect Mild/Moderate labels and links that are more than 30 days or a name match away from the echo. Once a model has learnt the patterns that generalise, the only way to raise training accuracy further is to memorise the individual training ECGs, including their noise and label errors. That memorised detail does not exist in new patients, so held-out accuracy cannot rise and often falls.

The best stopping point is therefore not 100 % training accuracy but the epoch at which accuracy on data the model has not seen stops improving. In this project that is the inner-validation early-stopping point used in all original runs, where training accuracy is about 0.60–0.85. The gap between training and held-out accuracy is the measure of over-fitting, and the held-out accuracy is the number to report.

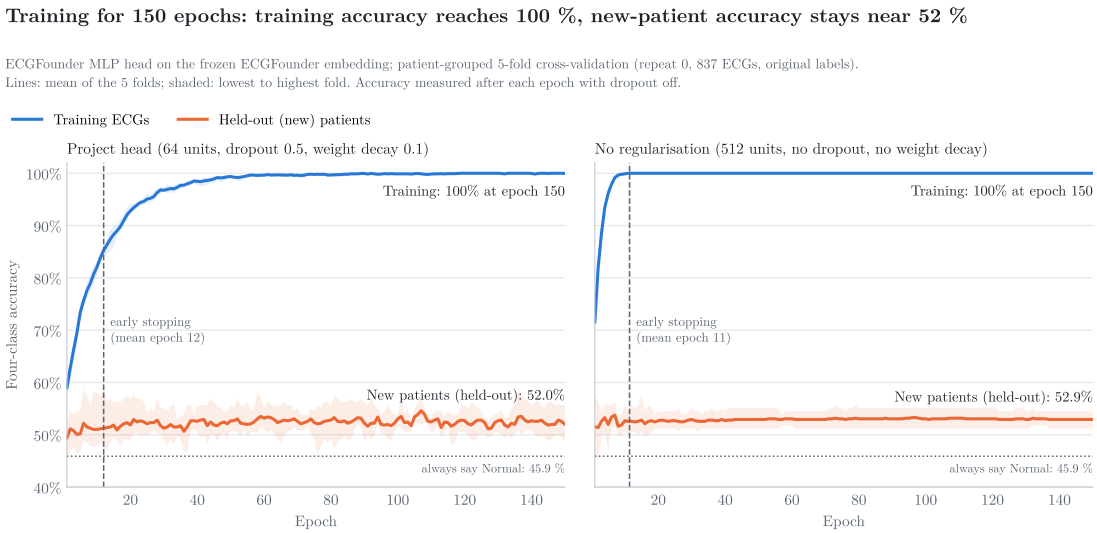


Figure 7.3 ECGFounder MLP head trained for 150 epochs without early stopping (mean of the five folds of repeat 0): training accuracy reaches 1.000 while held-out accuracy stays at about 0.52. Left: the project head; right: the same head without dropout or weight decay. The dashed line marks the early-stopping epoch.

Chapter 8

Summary and Conclusions

8.1 Conclusions

1. **Best on all patients: 0.587** [0.565, 0.610] from image-derived inputs only, about 99 % of the 0.592 that the EF information of the ECG permits, and above the model that also reads the printed measurements (0.562).
2. **Best in the 60s: 0.694** [0.655, 0.736] on the 56 % of patients the triage model answers in cross-validation.
3. **Best in the 70s: 0.747** [0.660, 0.837] on the 50 % of unseen locked-test patients it answers, above its own development estimate.
4. **Transfer works across populations:** a model trained only on US public data detects EF $\leq$ 45 % on our printouts with AUROC 0.85.
5. **Training to 100 % does not help:** no model retrained until it fitted its training data perfectly became reliably better on held-out patients, and most became worse (Chapter 7); the early-stopped models, with training accuracy of about 0.60–0.85, are the ones to use.
6. **The ceiling is set by the labels and the information in the ECG:** 70 % on all patients would need EF estimates correlating at $r \approx 0.90$ with echo, against about 0.66 achievable today.

8.2 Scope for future work (Phase II)

- Larger, multi-year data with biplane-Simpson EF linked by patient identifier within days of the ECG.
- Digital 10-s simultaneous ECGs where available, and enrichment for Mild and Moderate patients.
- Prospective evaluation of the triage model as an echocardiography-referral aid, and deployment as a lightweight application.

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